An Online Spectral Bundle Method for Large Scale SDP with Local Linear Convergence

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September 2026

Abstract

Classical spectral bundle method exploits low-rank structure, but their iterations typically require a dominant eigenspace computation and the solution of a reduced quadratic SDP subproblem, both of which can be computationally demanding. To reduce this computational burden, we propose an online spectral bundle method that replaces these two operations by one Riemannian gradient descent step for the Ky-Fan maximization problem corresponding to the eigenspace computation and one projected-gradient step for the reduced SDP subproblem respectively. Together with a damped dual update, these operations yield an iteration with no inner loops for either eigenspace computation or reduced optimization; Under standard SDP regularity conditions, we establish local linear convergence of both primal and dual iterates for suitable fixed parameters. The working rank may exceed the solution rank without requiring an additional spectral gap at the working rank.

1 Introduction

Semidefinite programming (SDP) provides convex models and relaxations for problems whose natural variables are matrices of pairwise interactions, moments, or lifted products. Representative examples include the Goemans–Williamson relaxation for combinatorial optimization [18] and the PhaseLift formulation of phase retrieval [9]. These applications illustrate the value of a convex formulation: primal and dual solutions provide bounds and certificates in addition to candidate solutions. They also expose a basic computational difficulty. An explicit symmetric matrix of order n requires quadratic storage, and repeatedly factoring matrices of this size can dominate the cost of an SDP algorithm. When n is sufficiently large to preclude dense storage, an implicit representation is essential. In problems with low-rank solutions, low-rank factors offer a natural way to meet this requirement; sparsity and other problem structure can provide complementary forms of compression.

A standard approach is the Burer–Monteiro (BM) factorization $X = UU^\top$, with $U \in \mathbb{R}^{n\times r}$ and $r \ll n$ [7, 8]. It enforces positive semidefiniteness through the representation and reduces matrix storage to $O(nr)$. The resulting nonlinear program can be treated by augmented Lagrangian methods or, when the constraints define a suitable smooth manifold, by Riemannian optimization. Rank-adaptive procedures use spectral optimality tests to decide when to enlarge the factor; the Riemannian staircase is a prominent example [4]. This approach has substantial theoretical support. Under appropriate regularity and sufficient factor width, generic landscape results connect second-order critical points of the factorized problem to global SDP solutions [5].

The computational benefit of factorization nevertheless comes with questions that depend on the chosen rank and formulation. A small factor width can create spurious local minima, even when the original SDP has a low-rank optimum [34]. Increasing the width raises storage and arithmetic costs,

while rank adaptation and optimality certification may themselves require spectral computations. Modern algorithms address these issues through different mechanisms: low-rank ADMM uses a bilinear splitting [20], ALORA exploits local structure of BM augmented-Lagrangian subproblems [14], and HALLaR combines convex and nonconvex steps with global complexity guarantees [28]. An important practical goal is to combine a small matrix representation, inexpensive updates, and a convergence guarantee appropriate to the original convex SDP.

An alternative is to retain the convex problem and its primal–dual interpretation while using low-rank atoms or reduced matrix models in the algorithm. The conditional-gradient-based augmented Lagrangian method (CGAL) is an important example [36]. Its primal update uses a linear minimization oracle, which reduces to an extreme-eigenvector computation on a PSD trace ball, together with a dual update. Its general guarantee for objective error and affine infeasibility is $O(k^{-1/2})$. This sublinear guarantee motivates methods with faster local convergence when high accuracy is needed. SketchyCGAL further reduces memory by maintaining a randomized sketch of the primal matrix [37]. These are conditional-gradient methods, rather than ordinary subgradient descent on the nonsmooth dual objective. Nor is sublinear convergence a limitation of the entire conditional-gradient family: spectral and other Frank–Wolfe variants have linear convergence guarantees under additional structure [12, 17].

Spectral bundle methods provide another convex route to low-rank computation [21]. They approximate the spectral dual objective by a model supported on a small subspace and compute a proximal step for that model. When the retained spectral information captures the active eigenspace, quadratic model accuracy can yield local linear convergence [13]. Inexact spectral bundle methods allow approximate spectral information [27], and the bundle-based augmented Lagrangian framework (BALA) makes the primal–dual relationship explicit [26]. These developments already establish that bundle models and linear convergence can coexist in a single outer loop. For spectral-subspace bundle constructions, the remaining work typically includes computing dominant spectral information and solving a reduced model problem. Warm starts, low-dimensional solvers, and sketching improve this cost, but an outer iteration need not consist of a fixed number of elementary updates.

This paper investigates a concrete way to replace the eigenspace solve and iterative reduced solve in this construction. We consider

$$\min\{\langle C, X\rangle : \ \mathcal{A}(X) = b, \ X \succeq 0, \ \operatorname{tr}(X) \le 1\},$$

where $C \in \mathbb{S}^n$, $\mathcal{A} : \mathbb{S}^n \to \mathbb{R}^m$ is linear, and $b \in \mathbb{R}^m$. The method maintains a dual variable y_k, an orthonormal matrix $V_k \in \mathbb{R}^{n\times r}$, and a PSD matrix X_k supported on $\operatorname{Range}(V_k)$. Each iteration first updates the dual variable using the current affine residual. It then takes one residual step for the subspace associated with $H(y) = \mathcal{A}^* y - C$, followed by a thin QR factorization. Finally, it transports X_k into the new subspace and takes one projected-gradient step for the reduced proximal target. We call this the *online spectral bundle method*: spectral and primal information are refined once as the dual iterate advances. Here "online" describes this interleaving of updates for a fixed SDP; the data need not arrive sequentially.

The computational statement is precise. There is no inner convergence test for an ambient eigenspace or for the reduced optimization problem. The two eigendecompositions in the iteration have order r, and there is one thin $n \times r$ QR factorization. In a factor implementation, $X_k = V_k W_k V_k^\top$ need not be formed as a dense matrix, so the state requires $O(nr + r^2 + m)$ storage, apart from the problem data and operator workspace. The cost of applying $\mathcal{A}$ and $\mathcal{A}^*$ still depends on the data representation. Throughout, "convex low-rank" refers to the original convex SDP and the convex reduced matrix model; the subspace variable itself ranges over a nonconvex set.

Our contribution is a convergence analysis for this particular combination of fixed updates.

The component subspace steps, warm-start strategies, and comparison-system arguments have precedents [38, 24, 16]. The issue addressed here is their simultaneous interaction with a moving reduced feasible set and an inexact reduced optimizer. The main results are as follows.

(i) *Local primal–dual linear convergence with one update of each state.* Under the standing SDP regularity assumptions and local dual quadratic growth, Theorem 5.20 proves that the primal and dual iterates converge linearly from a sufficiently close initialization. Theorem 5.19 supplies feasible fixed parameters, including a scaling with subspace step $\eta_V = \varepsilon$ and dual damping $\tau = \varepsilon^2$ for sufficiently small fixed $\varepsilon > 0$. The convergence theorem concerns the iterates of the stated algorithm, with exactly one reduced projected-gradient step per iteration.

(ii) *Uniform reduced-target control with redundant working directions.* If the selected primal solution has rank $s < r$, the working subspace contains $r - s$ directions that need not approach a unique rank-r eigenspace. We establish reduced-target regularity and quadratic growth uniformly near the entire family of rank-r projectors containing the optimal support (Theorem 5.11). The analysis uses the gap at rank s and does not impose an additional gap at rank r.

(iii) *A vanishing drift bound and a closed local comparison.* The reduced target changes when the working subspace changes. We show that the squared mismatch between its transported old value and its new value is bounded by the squared subspace stepsize η_V^2 times the current dual and spectral errors (Theorem E.33). This bound captures cancellation along the redundant projector family. It allows the dual, spectral, and reduced-solution errors to satisfy a contracting three-dimensional comparison system in an invariant neighborhood (Theorems 5.18 and 5.20).

These results concern the local regime of a fixed SDP. In particular, the regularity assumptions include an inactive trace bound at the selected solution, strict complementarity, primal and dual nondegeneracy, and local dual quadratic growth. We do not infer global convergence from arbitrary initialization or a wall-clock advantage from the local rate. The purpose is to establish that these two specific single-step replacements can preserve linear convergence.

Section 2 places the method among related algorithms and convergence frameworks. Section 3 introduces the SDP, its spectral dual model, and the reduced proximal target. Section 4 presents the algorithm, and Section 5 gives the convergence analysis. Detailed proofs appear in the appendices.

2 Literature review

Reduced models and inexact spectral bundles. Spectral bundle methods attain local linear convergence when their models retain sufficiently many active spectral directions [13, 25]; the working rank may already exceed the active rank. Their iterations retain a reduced optimization problem and leading-eigenspace computations. Lin [27] allows controlled spectral-oracle errors while solving a small quadratic SDP, and USBS [2] combines spectral bundling with sketching and warm starts while retaining iterative reduced optimization.

BALA [26] provides a closely related single-loop framework. Its linear theorem uses quadratic model closeness, which forces interpolation at the model center, and exact restricted minimization; its spectral-subspace SDP construction also uses newly computed dominant eigenvectors. Our model can have a nonzero center error, as reflected in (41), and the reduced matrix receives only one projected-gradient update. Convergence therefore requires controlling both evolving errors.

Low-rank subproblems and spectral tracking. ALORA [14] establishes related regularity results: localized augmented-Lagrangian subproblems inherit low rank, strict complementarity, and quadratic growth, enabling local linear convergence of gradient descent on a correctly sized factorization for a fixed subproblem. Our reduced target changes with both the dual center and the supporting subspace, so fixed-subproblem contraction alone is insufficient. Spectral Frank–Wolfe [12] also combines a small spectral subspace with corrective reduced optimization, but its smooth matrix-optimization setting does not directly provide the present primal–dual guarantee with general affine equalities.

The residual subspace step has precedents in block eigensolvers and QR-based subspace gradient methods [38, 16]. For polynomial-filtered proximal gradient, Li et al. [24] show that a sufficiently large constant filter degree can preserve linear convergence by coupling optimization and eigenspace errors. Here the reduced optimizer has an additional tracking error, is generally not a spectral function of $H(y)$, and belongs to a feasible set that moves with the subspace.

Single-step convergence frameworks. Relative-error proximal methods [31], approximate spectral projections in ADMM [30], and single-loop tracking schemes [33] provide mechanisms for controlling computational errors. Xu [35] proves local linear convergence with one primal proximal-gradient update under local assumptions including an interior primal solution; our rank-s target lies on the boundary of the rank-r reduced PSD cone when $s < r$. These results motivate the analysis but do not supply the estimates needed for our updates. We establish uniform reduced-target control near the entire family of subspaces containing the optimal support and a drift bound that vanishes with the dual and spectral errors. These estimates justify the specific combination of one dual, one subspace, and one reduced-matrix update.

3 Problem, spectral model, and reduced proximal target

Notation. All vector spaces are real. We write $\mathbb{S}^n$ for the space of symmetric $n \times n$ matrices and $\mathbb{S}^n_+$ for its positive semidefinite (PSD) cone; $X \succeq 0$ and $X \succ 0$ mean positive semidefiniteness and positive definiteness, respectively, and $X \succeq Y$ means $X - Y \succeq 0$. For matrices of the same size, the Frobenius inner product and norm are $\langle X, Y\rangle := \operatorname{tr}(X^\top Y)$ and $\|X\|_F := \sqrt{\langle X, X\rangle}$. We use $\|X\|_2$ for the spectral norm (the largest singular value), $\|X\|_*$ for the nuclear norm (the sum of singular values), and $\|x\|_2$ for the Euclidean norm of a vector. For a linear map $\mathcal{T}$, the induced norm is $\|\mathcal{T}\|_{a\leftarrow b} := \sup_{h\neq 0} \|\mathcal{T}h\|_a / \|h\|_b$. Product spaces carry the product Euclidean norm, with Frobenius norms on matrix components; D denotes the Fréchet derivative, and $\|\cdot\|_{\mathrm{op}}$ denotes the corresponding operator or multilinear norm. The symbols tr, rank, Range, and Null denote trace, rank, range, and nullspace. The identity matrix is I (or I_r when its size needs to be specified), and

$$\mathrm{St}(n, r) := \{V \in \mathbb{R}^{n\times r} : V^\top V = I_r\}, \qquad O(r) := \mathrm{St}(r, r).$$

For $H \in \mathbb{S}^n$, its eigenvalues are ordered as $\lambda_1(H) \geq \cdots \geq \lambda_n(H)$. For a nonempty closed convex set K in a Euclidean space and $x \in K$, we write

$$T_K(x) := \overline{\{t(z - x) : t \geq 0,\ z \in K\}}, \qquad \operatorname{lin} T_K(x) := T_K(x) \cap \big(-T_K(x)\big)$$

for its tangent cone and the lineality space of that cone, respectively; the latter is the largest linear subspace contained in $T_K(x)$. The symbols $\operatorname{aff} K$, $\operatorname{ri} K$, and $\operatorname{conv} K$ denote affine hull, relative interior, and convex hull. The map Proj_K is the metric projection onto K, using the Frobenius norm for matrices, and $\mathrm{dist}_F(X, K) := \inf_{Y\in K} \|X - Y\|_F$. We use $B_a(x)$ and $\overline{B}_a(x)$ for open and

closed Euclidean balls, and ∂f for the convex subdifferential of a convex function f. We use the standard Weyl, Ky Fan, Rayleigh–Ritz, principal-angle, and invariant-subspace perturbation results from [15, 11, 23, 32]; these basic linear-algebraic tools are not reproved.

Definition 3.1 (Trace-bounded SDP and spectral matrix)**.** Given $C, A_1, \dots, A_m \in \mathbb{S}^n$ and $b \in \mathbb{R}^m$, define

$$\mathcal{A}(X) := (\langle A_1, X\rangle, \dots, \langle A_m, X\rangle), \qquad \mathcal{A}^* y := \sum_{i=1}^m y_i A_i, \qquad H(y) := \mathcal{A}^* y - C.$$

The primal problem is

$$\begin{aligned} \min_{X \in \mathbb{S}^n} \quad & \langle C, X\rangle \\ \text{s.t.} \quad & \mathcal{A}(X) = b, \\ & X \succeq 0, \qquad \operatorname{tr} X \le 1. \end{aligned} \tag{1}$$

We write $\mathcal{D} := \{X \succeq 0 : \operatorname{tr} X \le 1\}$.

Lemma 3.2 (Linear optimization over the PSD trace ball)**.** *For every $G \in \mathbb{S}^n$,*

$$\max_{X \in \mathcal{D}} \langle G, X\rangle = \max\{0, \lambda_1(G)\}. \tag{2}$$

If $\lambda_1(G) > 0$, the maximizers are precisely the matrices in $\mathcal{D}$ with trace one whose ranges lie in the top eigenspace of G; if $\lambda_1(G) < 0$, the unique maximizer is zero.

Dual objective. Keeping the PSD and trace constraints in the domain $\mathcal{D}$, define the Lagrangian by

$$\mathcal{L}(X, y) := \langle C, X\rangle + y^\top \big(b - \mathcal{A}(X)\big), \qquad X \in \mathcal{D}, \quad y \in \mathbb{R}^m. \tag{3}$$

The primal and dual problems have the saddle formulations

$$\text{primal:} \quad \min_{X \in \mathcal{D}} \max_{y \in \mathbb{R}^m} \mathcal{L}(X, y), \qquad \text{dual:} \quad \max_{y \in \mathbb{R}^m} \min_{X \in \mathcal{D}} \mathcal{L}(X, y). \tag{4}$$

The inner maximization in the primal is understood in the extended-value sense: it equals $\langle C, X\rangle$ when $\mathcal{A}(X) = b$ and is $+\infty$ otherwise. By Lemma 3.2, the negative dual objective is the convex function

$$\begin{aligned} F(y) &:= \max_{X \in \mathcal{D}} \big[-\mathcal{L}(X, y)\big] \\ &= -b^\top y + \max_{X \in \mathcal{D}} \langle H(y), X\rangle = -b^\top y + \max\{0, \lambda_1(H(y))\}. \end{aligned} \tag{5}$$

Thus the dual problem is equivalently $\min_{y \in \mathbb{R}^m} F(y)$.

3.1 Standing KKT point and regularity assumptions

Assume that (1) satisfies Slater's condition: there exists $\widehat{X} \succ 0$ with $\mathcal{A}(\widehat{X}) = b$ and $\operatorname{tr} \widehat{X} < 1$. Let $(X_\star, y_\star, S_\star, \nu_\star)$ be a selected KKT point, so

$$\begin{aligned} &\mathcal{A}(X_\star) = b, \quad X_\star \succeq 0, \quad \operatorname{tr} X_\star \le 1, \\ &S_\star = C - \mathcal{A}^* y_\star + \nu_\star I \succeq 0, \quad \nu_\star \ge 0, \\ &S_\star X_\star = 0, \quad \nu_\star(1 - \operatorname{tr} X_\star) = 0. \end{aligned} \tag{6}$$

Here $S_\star$ is the PSD slack and $\nu_\star$ is the scalar multiplier for the trace constraint. The necessity and sufficiency of this system follow from the standard finite-dimensional convex KKT theorem under Slater's condition; see [29, 6]. Write

$$X_\star = U_\star \Sigma_\star U_\star^\top, \qquad U_\star \in \operatorname{St}(n, s), \quad \Sigma_\star \succ 0, \qquad P_\star := U_\star U_\star^\top. \tag{7}$$

Assumption 3.3 (Standard SDP regularity package)**.** *The selected KKT point satisfies*

(S1) trace inactivity: $\operatorname{tr} X_\star < 1$*;*

(S2) PSD strict complementarity: $\operatorname{rank} X_\star + \operatorname{rank} S_\star = n$*;*

(S3) primal nondegeneracy: $\mathcal{A}(\operatorname{lin} T_{\mathbb{S}^n_+}(X_\star)) = \mathbb{R}^m$*;*

(S4) dual nondegeneracy: $\mathcal{A}^*(\mathbb{R}^m) + \operatorname{lin} T_{\mathbb{S}^n_+}(S_\star) = \mathbb{S}^n$.

The integers $1 \le s < r \le n$ *are fixed.*

This terminology agrees with the SDP nondegeneracy framework of Alizadeh, Haeberly, and Overton [1]. Only (S1), (S2), and (S4) are used in the reduced-target construction below: (S1) removes the scalar trace multiplier, (S2) separates the primal and slack supports, and (S4) supplies active-block injectivity. We retain (S3) because it belongs to the standard two-sided regularity package, but we do not silently use it.

3.2 Spectral model and reduced proximal target

Definition 3.4 (Reduced spaces)**.** For $1 \le r \le n$, let

$$\Theta_r := \{W \in \mathbb{S}^r : W \succeq 0,\ \operatorname{tr} W \le 1\}.$$

For $V \in \operatorname{St}(n,r)$ and $P = VV^\top$, set

$$\mathcal{E}(P) := \{X \in \mathbb{S}^n : PXP = X\}, \qquad \mathcal{D}(P) := \mathcal{D} \cap \mathcal{E}(P).$$

We also write

$$\mathfrak{P}_r := \{P \in \mathbb{S}^n : P^\top = P,\ P^2 = P,\ \operatorname{tr} P = r\}. \tag{8}$$

Every $P \in \mathfrak{P}_r$ has a representation $P = VV^\top$ with $V \in \operatorname{St}(n,r)$, and conversely every such product belongs to $\mathfrak{P}_r$.

Lemma 3.5 (Coefficient-to-matrix correspondence)**.** *Let* $V \in \operatorname{St}(n,r)$ *and* $P = VV^\top$*. Then*

$$\mathcal{D}(P) = \{VWV^\top : W \in \Theta_r\}, \qquad \mathcal{E}(P) = \{VWV^\top : W \in \mathbb{S}^r\}.$$

Definition 3.6 (Spectral model and exact reduced target)**.** For $V \in \operatorname{St}(n,r)$, define the full spectral-bundle model

$$M_V(z) := -b^\top z + \max_{W \in \Theta_r} \left\langle V^\top H(z) V, W \right\rangle. \tag{9}$$

It is a global minorant of F. For $\rho > 0$, $y \in \mathbb{R}^m$, and $P = VV^\top$, define

$$\Psi_{\rho,y}(X) := \frac{1}{2}\|\mathcal{A}(X) - b\|_2^2 - \rho \langle H(y), X\rangle, \qquad X \in \mathcal{D}(P). \tag{10}$$

Whenever the minimizer is unique, it is denoted by $X^\sharp_\rho(y,P)$ and called the exact reduced target.

Proposition 3.7 (Model-prox/reduced-SDP equivalence)**.** *Let* $y \in \mathbb{R}^m$*,* $\rho > 0$*,* $V \in \operatorname{St}(n,r)$*, and* $P = VV^\top$*. Let*

$$z^+ \in \operatorname*{argmin}_{z \in \mathbb{R}^m} \left\{ M_V(z) + \frac{\rho}{2}\|z - y\|_2^2 \right\} \tag{11}$$

and

$$X^+ \in \operatorname*{argmin}_{X \in \mathcal{D}(P)} \Psi_{\rho,y}(X). \tag{12}$$

Then z^+ is unique and

$$z^+ = y - \frac{1}{\rho}\big(\mathcal{A}(X^+) - b\big). \tag{13}$$

A direct proof of this equivalence is recorded in Section A; see also Helmberg and Rendl [21].

From classical spectral bundles to an online method. Direct proximal descent on F would require solving a large-scale nonsmooth spectral optimization problem at every iteration, which is generally too costly in practice. The classical spectral bundle idea is to approximate F by the simpler minorant M_V and take a proximal step on that model; see [21]. For the local comparison used here, we isolate the exact-eigenspace iteration

$$\begin{aligned} V_k &\in \operatorname*{argmax}_{V \in \mathrm{St}(n,r)} \operatorname{tr}\big(V^\top H(y_k) V\big), \\ y_{k+1} &\in \operatorname*{argmin}_{y \in \mathbb{R}^m} \left\{ M_{V_k}(y) + \frac{\rho}{2}\|y - y_k\|_2^2 \right\}. \end{aligned} \tag{14}$$

The full classical method also includes bundle management and a serious/null-step acceptance test. In (14), the first line computes a top-r eigenspace of $H(y_k)$. By Proposition 3.7, the second line is computed through

$$X_k^{\mathrm{red}} \in \operatorname*{argmin}_{X \in \mathcal{D}(P_k)} \Psi_{\rho, y_k}(X), \qquad y_{k+1} = y_k - \frac{1}{\rho}\big(\mathcal{A}(X_k^{\mathrm{red}}) - b\big), \quad P_k = V_k V_k^\top. \tag{15}$$

Near a regular solution, a model containing the active spectral subspace of $H(y_k)$ has a second-order approximation error:

$$0 \le F(y) - M_{V_k}(y) \le \frac{\beta}{2}\|y - y_k\|_2^2 \qquad \text{for } y \text{ and } y_k \text{ sufficiently close to } y_\star. \tag{16}$$

In our setting this is the zero-spectral-error case of Lemma 5.9. Together with local quadratic growth of F and a sufficiently large proximal parameter $\rho > \beta$, this estimate makes the exact model-proximal iteration locally linearly convergent. This second-order model-accuracy mechanism also underlies the local linear convergence theory of classical spectral bundle methods; see Ding and Grimmer [13].

Despite the reduced model, each classical iteration still computes a dominant eigenspace of the large matrix $H(y_k)$ and uses an inner solver to solve the reduced optimization problem. Our goal is to preserve local linear convergence while replacing both computations by one inexpensive update per iteration. The resulting *online spectral bundle method* maintains and updates a subspace and a reduced primal matrix as the dual variable evolves.

4 Online spectral bundle algorithm

Design motivation. To implement the replacement described at the end of Section 3, the online spectral bundle method keeps the reduced primal matrix X_k and the history basis V_k as evolving states. One full-residual Grassmann step updates V_k, and one projected-gradient step updates X_k on the new reduced feasible set. A relaxed dual step uses the current primal residual. Thus the subspace and primal states track targets that move with the iteration, and the analysis must show that both tracking errors contract jointly with the dual error.

At iteration k, the method stores a dual vector y_k, an orthonormal history basis $V_k \in \mathrm{St}(n,r)$, and a reduced primal matrix $X_k \in \mathcal{D}(P_k)$, where $P_k = V_k V_k^\top$. Each of the three states is updated once; there is no inner loop for either the reduced SDP or the eigenspace computation. We use standard symmetric eigendecompositions and thin QR factorizations; see, e.g., [19].

Algorithm 1. Online spectral bundle method

Input: $y_0 \in \mathbb{R}^m$, $V_0 \in \mathrm{St}(n,r)$, $X_0 \in \mathcal{D}(P_0)$; fixed $\rho, \tau, \eta_V, \eta_X > 0$.

1. For $k = 0, 1, 2, \ldots$, set $y_{k+1} = y_k - \dfrac{\tau}{\rho}(\mathcal{A}(X_k) - b)$.
2. Set $H_{k+1} = H(y_{k+1})$ and compute $V_k^\top H_{k+1} V_k = Y_k \,\mathrm{diag}(\mu_1^k, \ldots, \mu_r^k) Y_k^\top$ with $\mu_1^k \geq \cdots \geq \mu_r^k$.
3. Set $\widetilde{V}_k = V_k Y_k$ and $D_k = (I - \widetilde{V}_k \widetilde{V}_k^\top) H_{k+1} \widetilde{V}_k$.
4. Compute one thin QR factorization without column pivoting: $\widetilde{V}_k + \eta_V D_k = V_{k+1} R_k^{\mathrm{QR}}$, with $\mathrm{diag}(R_k^{\mathrm{QR}}) > 0$.
5. Set $\widehat{X}_{k+1} = \mathrm{Proj}_{\mathcal{D}(P_{k+1})}(X_k)$.
6. Set $X_{k+1} = \mathrm{Proj}_{\mathcal{D}(P_{k+1})}[\widehat{X}_{k+1} - \eta_X \nabla \Psi_{\rho, y_{k+1}}(\widehat{X}_{k+1})]$.

Lemma 4.1 (PSD compression is Frobenius projection)**.** *If $P \in \mathfrak{P}_r$, $X \succeq 0$, and $\mathrm{tr}\, X \leq 1$, then $PXP \in \mathcal{D}(P)$ and*

$$\mathrm{Proj}_{\mathcal{D}(P)}(X) = PXP. \tag{17}$$

Corollary 4.2 (Reduced trace-ball projection uses only an $r \times r$ eigendecomposition)**.** *Let $P = VV^\top \in \mathfrak{P}_r$ and $X \in \mathbb{S}^n$. Then*

$$\mathrm{Proj}_{\mathcal{D}(P)}(X) = V \,\mathrm{Proj}_{\Theta_r}(V^\top X V)\, V^\top. \tag{18}$$

If $V^\top X V = Q \,\mathrm{diag}(\lambda) Q^\top$, then

$$\mathrm{Proj}_{\Theta_r}(V^\top X V) = Q \,\mathrm{diag}(w) Q^\top, \qquad w = \mathrm{Proj}_{\{u \geq 0:\, \mathbf{1}^\top u \leq 1\}}(\lambda). \tag{19}$$

By Lemma 4.1 and Corollary 4.2, the projection in Step 5 of Algorithm 1 is computed as $P_{k+1} X_k P_{k+1}$, while the projection in Step 6 is computed by compressing its argument with V_{k+1}, performing an $r \times r$ eigendecomposition and a capped-simplex projection, and lifting the result back with V_{k+1}. The gradient used in Step 6 is

$$\nabla \Psi_{\rho, y}(X) = \mathcal{A}^*(\mathcal{A}(X) - b) - \rho H(y). \tag{20}$$

Definition 4.3 (Intrinsic full-residual projector update)**.** For $y \in \mathbb{R}^m$, $V \in \mathrm{St}(n,r)$, $P = VV^\top$, and $\eta \in \mathbb{R}$, define

$$\begin{aligned} D(y, V) &:= (I - P) H(y) V, \\ P^+(y, P, \eta) &:= Z(Z^\top Z)^{-1} Z^\top, \qquad Z := V + \eta D(y, V). \end{aligned} \tag{21}$$

Lemma 4.4 (Well-definedness and basis invariance of P^+)**.** *Let $y \in \mathbb{R}^m$, $P \in \mathfrak{P}_r$, and $\eta \in \mathbb{R}$. Then $P^+(y, P, \eta)$ is a rank-r orthogonal projector and is independent of the particular orthonormal basis V of* $\mathrm{Range}(P)$*. Moreover, the update in Algorithm 1 satisfies*

$$P_{k+1} = P^+(y_{k+1}, P_k, \eta_V). \tag{22}$$

Starting from the feasible input in Algorithm 1, every algorithmic state is well defined: the small symmetric eigendecomposition exists, Lemma 4.4 gives full column rank for the QR input, and both projections are onto nonempty compact convex sets and are unique. Consequently, $V_k \in \mathrm{St}(n, r)$ and $X_k \in \mathcal{D}(P_k)$ for every $k \geq 0$.

5 Convergence analysis: roadmap and main mechanisms

Proximal-parameter interval. For the convergence analysis, fix a compact proximal-parameter interval

$$0 < \rho_- < \rho_+ < \infty. \tag{23}$$

All reduced-target neighborhoods and constants below are constructed uniformly for $\rho \in [\rho_-, \rho_+]$. A single actual value of ρ is selected from this interval before an algorithmic run and is then kept fixed. The stronger lower bound needed for the dual recursion will be imposed only after the spectral-model constants have been constructed.

Proof roadmap. All proofs are deferred to the appendices. The purpose of this section is to expose the mechanisms that drive the analysis and how they fit together. With $P_k = V_k V_k^\top$, and whenever the exact reduced target is locally single-valued, the three state errors are

$$d_k = \|y_k - y_\star\|_2^2, \qquad \sigma_k = \Phi_s(H(y_k), V_k), \qquad \xi_k = \left\| X_k - X_\rho^\sharp(y_k, P_k) \right\|_F^2.$$

They measure, respectively, dual displacement, active spectral-subspace error, and failure of the single projected-gradient iterate to reach the current reduced minimizer. Here Φ_s is defined in Definition 5.3, and $X_\rho^\sharp(y_k, P_k)$ is defined in Definition 3.6; its required local existence and uniqueness are established in Section 5.3. The proof is organized around four points.

(i) *Spectral-subspace contraction.* The full-residual direction is a Grassmann–Riemannian gradient direction. Near the solution, the active spectral gap makes one such step contract σ_k.

(ii) *Dual proximal contraction.* Proximal descent and dual quadratic growth contract d_k.

(iii) *Primal reduced-target contraction.* Projected-gradient descent and uniform quadratic growth of the reduced objective contract ξ_k.

(iv) *Coupling and target drift.* Updating y_k moves the active eigenspace and the reduced target, while updating V_k changes the reduced feasible set. These drift errors must remain smaller than the three contractions. Section 5.4 collects all effects in a positive 3×3 comparison system and proves its local stability.

5.1 Preparation for the spectral-subspace contraction

Strict complementarity separates the $s = \operatorname{rank} X_\star$ active eigenvalues from the remaining spectrum. Because no gap is assumed at index r, the analysis must track an s-dimensional subspace hidden inside the r-dimensional history space. The next two lemmas provide the pointwise gap and its uniform local continuation.

Lemma 5.1 (Strict complementarity gives the hidden active gap)**.** *Under (S1)–(S2),*

$$\mathrm{Null}(S_\star) = \mathrm{Range}(U_\star), \qquad S_\star \succeq \delta_\star (I - P_\star) \tag{24}$$

for some $\delta_\star > 0$. Consequently,

$$\lambda_s(H(y_\star)) = 0, \qquad \lambda_{s+1}(H(y_\star)) \le -\delta_\star, \qquad \lambda_s(H(y_\star)) - \lambda_{s+1}(H(y_\star)) \ge \delta_\star. \tag{25}$$

Lemma 5.2 (Uniform active gap, spectral width, and projector variation)**.** *Let*

$$\Delta_\star := \lambda_s(H(y_\star)) - \lambda_{s+1}(H(y_\star)) \ge \delta_\star, \qquad L_H := \|\mathcal{A}^*\|_{2\leftarrow 2}\,. \tag{26}$$

For a symmetric matrix whose sth spectral gap is positive, let $P_s(H)$ denote its unique leading rank-s spectral projector. Choose once and for all a radius $R_Y > 0$ satisfying

$$R_Y \le \frac{\Delta_\star}{16 L_H} \tag{27}$$

when $L_H > 0$; when $L_H = 0$, this restriction is omitted. Define

$$\mathcal{Y} := \{y \in \mathbb{R}^m : \|y - y_\star\|_2 \le R_Y\}, \qquad \underline{\Delta} := \frac{7}{8}\Delta_\star, \tag{28}$$

and

$$\overline{\Gamma} := \max_{y\in\mathcal{Y}} \big(\lambda_1(H(y)) - \lambda_n(H(y))\big), \qquad L_P := \frac{2\sqrt{2s}\, L_H}{\underline{\Delta}}. \tag{29}$$

Then $\overline{\Gamma} < \infty$ and, for every $y, z \in \mathcal{Y}$,

$$\begin{aligned} \lambda_s(H(y)) - \lambda_{s+1}(H(y)) &\ge \underline{\Delta}, \\ \lambda_1(H(y)) - \lambda_n(H(y)) &\le \overline{\Gamma}, \\ \|P_s(H(y)) - P_s(H(z))\|_F &\le L_P \|y - z\|_2\,. \end{aligned} \tag{30}$$

Lemma 5.1 identifies the active primal support with the kernel of the optimal slack. Lemma 5.2 extends the resulting eigengap to one dual neighborhood and makes the leading rank-s projector Lipschitz there. These are the common spectral constants used by every later tracking estimate.

Definition 5.3 (Hidden restricted Ky Fan subspace, gap, and residual)**.** For $H \in \mathbb{S}^n$, write

$$\Delta(H) := \lambda_s(H) - \lambda_{s+1}(H), \qquad \Gamma(H) := \lambda_1(H) - \lambda_n(H).$$

For $V \in \mathrm{St}(n, r)$, choose

$$Q_s(H, V) \in \operatorname*{argmax}_{\substack{Q\in\mathrm{St}(n,s)\\ \mathrm{Range}(Q)\subseteq\mathrm{Range}(V)}} \operatorname{tr}(Q^\top H Q)$$

and choose $U_s(H)$ as a global Ky Fan maximizer. When $\Delta(H) > 0$, set

$$P_s(H) := U_s(H)U_s(H)^\top, \qquad e_s(H, V) := \|(I - P_s(H))Q_s(H, V)\|_F^2\,,$$

and

$$\mathcal{R}_s(H, V) := (I - VV^\top)HQ_s(H, V). \tag{31}$$

The hidden Ritz gap is the basis-invariant quantity

$$\Phi_s(H, V) = \operatorname{tr}(U_s(H)^\top H U_s(H)) - \operatorname{tr}(Q_s(H, V)^\top H Q_s(H, V)).$$

Write $E_s(H) := \mathrm{Range}(U_s(H))$. Among all s-dimensional subspaces contained in $\mathrm{Range}(V)$, $\mathrm{Range}(Q_s(H,V))$ is closest to $E_s(H)$ in the sense of the Ky Fan objective value. The classical second-order spectral-bundle model condition is $E_s(H(y_k)) \subseteq \mathrm{Range}(V_k)$; the closer $\mathrm{Range}(Q_s(H(y_k), V_k))$ is to $E_s(H(y_k))$, the closer this inclusion is to being satisfied. The hidden Ritz gap Φ_s measures this discrepancy in Ky Fan objective value, while e_s measures it geometrically through principal angles. We track this hidden s-space because the spectral gap controls it, whereas the other $r - s$ history directions are redundant. The precise relations among these quantities are proved in Section C.

Proposition 5.4 (Compressed representation of the restricted Ky Fan problem)**.** *Let $H \in \mathbb{S}^n$, $V \in \mathrm{St}(n,r)$, and $B = V^\top H V$, and let $\mu_1 \geq \cdots \geq \mu_r$ be the eigenvalues of B. One may choose*

$$Q_s(H,V) = VY_s,$$

where $Y_s \in \mathrm{St}(r,s)$ is a leading orthonormal eigenbasis of B. For every choice of restricted Ky Fan maximizer,

$$\Phi_s(H,V) = \sum_{i=1}^{s} \lambda_i(H) - \sum_{i=1}^{s} \mu_i.$$

Proposition 5.4 explains the Ritz rotation in Steps 2–3 of Algorithm 1: the small decomposition of $V_k^\top H_{k+1} V_k$ exposes $Q_s(H_{k+1}, V_k)$ inside the history space. The subsequent full-residual step therefore updates the spectrally identifiable part of V_k without a large eigendecomposition.

Lemma 5.5 (Motion of the hidden Ritz gap)**.** *Set*

$$C_0 := \sqrt{2s}\, L_P L_H, \qquad C_1 := 2L_H \sqrt{\frac{s}{\underline{\Delta}}}. \tag{32}$$

For every $V \in \mathrm{St}(n,r)$ and $y, z \in \mathcal{Y}$,

$$\begin{aligned} \Phi_s(H(z),V) \leq \Phi_s(H(y),V) &+ C_1 \|z-y\|_2 \sqrt{\Phi_s(H(y),V)} \\ &+ C_0 \|z-y\|_2^2 . \end{aligned} \tag{33}$$

The dual update moves the target eigenspace before the residual step is taken. Lemma 5.5 bounds the Ky Fan tracking error created by this change from $H(y_k)$ to $H(y_{k+1})$.

Theorem 5.6 (One full-residual QR contraction)**.** *Let $H \in \mathbb{S}^n$, $V \in \mathrm{St}(n,r)$, and $1 \leq s < r$. If*

$$\Delta(H) > 0, \qquad \Gamma(H) > 0, \qquad \Phi_s(H,V) \leq \frac{\Delta(H)^2}{2\Gamma(H)}, \qquad 0 < \eta_V \Gamma(H) \leq 1,$$

then the Ritz rotation and full-residual QR update in Algorithm 1 produce V^+ satisfying

$$\Phi_s(H,V^+) \leq \left(1 - \eta_V \frac{\Delta(H)^2}{5\Gamma(H)}\right) \Phi_s(H,V). \tag{34}$$

In particular, set

$$\kappa_V := \frac{\underline{\Delta}^2}{5\overline{\Gamma}}. \tag{35}$$

For $H = H(y)$ with $y \in \mathcal{Y}$, if $0 < \eta_V \leq \overline{\Gamma}^{-1}$ and the same threshold holds, then

$$\Phi_s(H,V^+) \leq (1 - \kappa_V \eta_V) \Phi_s(H,V). \tag{36}$$

Theorem 5.6 is the basic contraction mechanism for σ_k: at a fixed matrix with an active eigengap, one full-residual QR step reduces the hidden Ritz gap. Lemma 5.5 supplies the separate control needed when the matrix changes between iterations.

5.2 Preparation for the dual proximal contraction

The dual mechanism combines an accurate local spectral model with quadratic growth of the dual objective.

Assumption 5.7 (Dual quadratic growth)**.** *There are $r_F > 0$ and $\mu_y > 0$ such that*

$$F(y) - F(y_\star) \geq \frac{\mu_y}{2} \|y - y_\star\|_2^2 \qquad \text{whenever } \|y - y_\star\|_2 \leq r_F. \tag{37}$$

Assumption 5.7 converts objective decrease into contraction of the dual distance. Since it is local, the exact model-proximal point must first be placed in its neighborhood.

Lemma 5.8 (Global model bounds)**.** *Let*

$$L_A := \|\mathcal{A}\|_{2\leftarrow F}, \qquad G_M := L_A + \|b\|_2. \tag{38}$$

Let $z_\rho(y, V)$ denote the unique minimizer in (11). Then, for every $V \in \mathrm{St}(n, r)$, $z, z', y \in \mathbb{R}^m$, and $\rho > 0$,

$$\left|M_V(z) - M_V(z')\right| \leq G_M \left\|z - z'\right\|_2, \qquad \|z_\rho(y, V) - y\|_2 \leq \frac{G_M}{\rho}. \tag{39}$$

Lemma 5.8 supplies this entry guarantee: for sufficiently large ρ, the exact model-proximal point stays inside both the spectral and quadratic-growth neighborhoods.

Lemma 5.9 (Hidden Ritz gap controls the spectral-model error)**.** *Set*

$$c_\Phi := 2, \qquad \beta := 2C_0 + \frac{C_1^2}{2} = \frac{10 s L_H^2}{\underline{\Delta}}. \tag{40}$$

For every $V \in \mathrm{St}(n, r)$ and $y, z \in \mathcal{Y}$,

$$0 \leq F(z) - M_V(z) \leq \frac{\beta}{2} \|z - y\|_2^2 + c_\Phi \Phi_s(H(y), V). \tag{41}$$

Lemma 5.9 links the spectral and dual mechanisms. It bounds the model error by a quadratic displacement term and the hidden Ritz gap. When $\Phi_s = 0$, this becomes the classical second-order spectral-bundle estimate. Together with proximal descent and Assumption 5.7, it yields the contraction mechanism for d_k.

5.3 Preparation for the primal reduced-target contraction

Throughout this subsection and Section E, Assumption 3.3 is in force. The reduced minimizer moves with both the dual center y and the history projector P. This subsection states the geometry and the conclusions needed in the convergence argument; the complete regularity theory is in Section E.

Uniform-dependence claim. Fix the SDP data $(C, \mathcal{A}, b)$, the selected KKT point $(X_\star, y_\star, S_\star)$, the ranks r, s, and the interval endpoints ρ_-, ρ_+. The complete list of reduced-target constants used in this and later subsections is

$$\begin{aligned} &\delta_\star, \gamma_\star, \delta_{\mathrm{chart}}, \eta_{\mathrm{chart}}, L_{X,1}, L_{X,11}, L_{Z,1}, L_{Z,11}, M_{P^+}, L_G, \\ &\qquad r_{\mathrm{trg}}, \underline{\sigma}, \overline{\sigma}, \underline{\nu}, \underline{\gamma}, \mu_X, R_Y, L_P, \underline{\Delta}, \\ &\qquad C_U, C_P, r_{\mathrm{track}}, r_{\mathrm{drift}}, \bar{\eta}_{\mathrm{drift}}, C_T. \end{aligned} \tag{42}$$

These constants may depend only on the listed fixed objects. They do not depend on an iteration index, the actual $\rho \in [\rho_-, \rho_+]$, or the algorithmic parameters τ, η_V, η_X. This claim is the standing dependence rule for the whole reduced-target construction.

Definition 5.10 (Redundant base-projector family)**.** Define

$$\mathfrak{P}_\star := \{P \in \mathfrak{P}_r : PP_\star = P_\star\}. \tag{43}$$

Thus $P \in \mathfrak{P}_\star$ if and only if $\mathrm{Range}(U_\star) \subseteq \mathrm{Range}(P)$.

Because $r > s$, the $r - s$ inactive directions have no distinguished limit. Every projector in $\mathfrak{P}_\star$ contains the active support and gives the same base reduced target $X_\star$. The local regularity estimates must therefore be uniform over this entire family.

Theorem 5.11 (Reduced-target regularity, tracking, and drift: summary)**.** *Under Assumption 3.3, there are positive constants*

$$r_{\mathrm{trg}},\ \mu_X,\ r_{\mathrm{track}},\ r_{\mathrm{drift}},\ \bar{\eta}_{\mathrm{drift}},\ C_P,\ L_{X,1},$$
$$0 < \underline{\sigma} \le \overline{\sigma} < \infty, \qquad \underline{\nu} > 0,$$

and a finite constant $C_T \ge 0$ such that, for every $\rho \in [\rho_-, \rho_+]$, the following properties hold.

(i) If $P \in \mathfrak{P}_r$, $\|y - y_\star\|_2 < r_{\mathrm{trg}}$, and $\mathrm{dist}_F(P, \mathfrak{P}_\star) < r_{\mathrm{trg}}$, then $X^\sharp_\rho(y, P)$ is the unique reduced minimizer. It and its associated reduced KKT slack $Z^\sharp_\rho(y, P)$ have complementary ranks s and $r - s$ and admit factorizations

$$X^\sharp_\rho(y, P) = U\Sigma U^\top, \qquad Z^\sharp_\rho(y, P) = R_c N R_c^\top, \tag{44}$$

where $[U, R_c] \in \mathrm{St}(n, r)$ and

$$\underline{\sigma} I \preceq \Sigma \preceq \overline{\sigma} I, \qquad N \succeq \underline{\nu} I. \tag{45}$$

(ii) For every triple in part (i) and every $X \in \mathcal{D}(P)$,

$$\Psi_{\rho,y}(X) - \Psi_{\rho,y}(X^\sharp_\rho(y, P)) \ge \frac{\mu_X}{2} \left\| X - X^\sharp_\rho(y, P) \right\|_F^2. \tag{46}$$

(iii) If $V \in \mathrm{St}(n, r)$, $P = VV^\top$, and

$$\|y - y_\star\|_2 + \sqrt{\Phi_s(H(y), V)} \le r_{\mathrm{track}}, \tag{47}$$

then the exact target is well defined and

$$\left\| X^\sharp_\rho(y, P) - X_\star \right\|_F \le L_{X,1}(1 + C_P)\big(\|y - y_\star\|_2 + \sqrt{\Phi_s(H(y), V)}\big). \tag{48}$$

(iv) If $V \in \mathrm{St}(n, r)$, $P = VV^\top$, $\|y - y_\star\|_2 \le r_{\mathrm{drift}}$, $\Phi_s(H(y), V) \le r^2_{\mathrm{drift}}$, and $0 \le \eta \le \bar{\eta}_{\mathrm{drift}}$, then both targets below are well defined and, for $P^+ = P^+(y, P, \eta)$,

$$\left\| P^+ X^\sharp_\rho(y, P) P^+ - X^\sharp_\rho(y, P^+) \right\|_F^2 \le C_T \eta^2 \big(\|y - y_\star\|_2^2 + \Phi_s(H(y), V)\big). \tag{49}$$

The complete independent statements are Corollary E.23 and Theorems E.27, E.31 and E.33 in Section E, and later derivations cite them directly. The complementary factorizations (44) and spectral bounds (45) describe the nearby reduced solutions uniformly over the redundant projector family. They provide the regularity needed to follow the reduced minimizer as the algorithm changes y and P.

Uniform reduced quadratic growth (46) holds on the entire reduced trace ball $\mathcal{D}(P)$ at each admissible nearby parameter triple. Consequently, the projected-gradient step that updates X in Algorithm 1 contracts toward the fixed reduced target from any feasible predictor; the precise contraction factor is given in (51). No separate closeness condition on that predictor is needed.

The tracking estimate (48) ties the exact reduced target to $X_\star$: improving the dual variable and the hidden spectral subspace brings this target closer to the original SDP solution. The drift estimate (49) controls a different change. When the algorithm updates V_k, both the reduced feasible set and its minimizer move. This estimate bounds the mismatch between the old target compressed into the new subspace and the new target. Its product form makes this mismatch smaller as the dual and spectral errors decrease, allowing the algorithm's single X update to retain its contraction toward a moving target.

Corollary 5.12 (One-X contraction at a fixed reduced target)**.** *Let (ρ, y, P) satisfy the parameter conditions of Corollary E.23, let $\widehat{X} \in \mathcal{D}(P)$, and let $0 < \eta_X \le L_A^{-2}$. Define*

$$X^+ := \mathrm{Proj}_{\mathcal{D}(P)}\Big(\widehat{X} - \eta_X \nabla \Psi_{\rho,y}(\widehat{X})\Big). \tag{50}$$

Then

$$\left\|X^+ - X_\rho^\sharp(y,P)\right\|_F^2 \le \frac{1}{1+\mu_X\eta_X}\left\|\widehat{X} - X_\rho^\sharp(y,P)\right\|_F^2. \tag{51}$$

The contraction (51) is the basic mechanism for reducing ξ_k when the algorithm's X update is applied at a fixed reduced target. The drift estimate (49) supplies the additional control required when the subspace changes.

5.4 The coupled recurrences and local linear convergence

Assumptions 3.3 and 5.7 are in force throughout this subsection and its proofs in Section F. All states considered below are generated by Algorithm 1; in particular, $V_k \in \mathrm{St}(n,r)$, $P_k = V_k V_k^\top$, and $X_k \in \mathcal{D}(P_k)$.

We now combine the three contractions and their drift terms. The lower bound on ρ_- keeps the exact model-proximal point inside the local spectral and quadratic-growth neighborhoods.

For this section and the remainder of the paper, fix the compact proximal-parameter box $[\rho_-, \rho_+]$ from (23) so that

$$\rho_- > \max\left\{\beta, \frac{2G_M}{\min\{R_Y, r_F\}}\right\}, \tag{52}$$

and fix one actual $\rho \in [\rho_-, \rho_+]$ for the entire run.

Definition 5.13 (The three state errors)**.** Whenever the exact reduced target is single-valued, abbreviate $X_k^\sharp := X_\rho^\sharp(y_k, P_k)$ and define

$$\begin{aligned} d_k &:= \|y_k - y_\star\|_2^2, \\ \sigma_k &:= \Phi_s(H(y_k), V_k), \\ \xi_k &:= \left\|X_k - X_k^\sharp\right\|_F^2. \end{aligned} \tag{53}$$

Define the exact model-proximal point associated with the current reduced target by

$$\widehat{y}_{k+1} := y_k - \frac{1}{\rho}\big(\mathcal{A}(X_k^\sharp) - b\big). \tag{54}$$

Proposition 3.7 shows that this is the unique minimizer of $M_{V_k}(z) + (\rho/2)\,\|z - y_k\|_2^2$.

Lemma 5.14 (Exact model-prox contraction and displacement)**.** *Let $\rho \in [\rho_-, \rho_+]$ and suppose that $X_k^\sharp = X_\rho^\sharp(y_k, P_k)$ exists uniquely. Assume*

$$\|y_k - y_\star\|_2 < r_{\mathrm{trg}}, \qquad \mathrm{dist}_F(P_k, \mathfrak{P}_\star) < r_{\mathrm{trg}}, \tag{55}$$

and

$$y_k, \widehat{y}_{k+1} \in \mathcal{Y}, \qquad \|\widehat{y}_{k+1} - y_\star\|_2 \le r_F. \tag{56}$$

Then

$$\|\widehat{y}_{k+1} - y_\star\|_2^2 \le \frac{\rho}{\rho + \mu_y} d_k + \frac{2c_\Phi}{\rho + \mu_y} \sigma_k, \tag{57}$$

$$\|\widehat{y}_{k+1} - y_k\|_2^2 \le \frac{\rho}{\rho - \beta} d_k + \frac{2c_\Phi}{\rho - \beta} \sigma_k. \tag{58}$$

Lemma 5.14 gives the ideal model-proximal contraction and its displacement. Replacing the exact target $X_k^\sharp$ by the maintained state X_k introduces the primal tracking error in the next recurrence.

Lemma 5.15 (The first quantitative recurrence)**.** *Define*

$$a_d := \frac{2\rho}{\rho - \beta}, \qquad a_\sigma := \frac{4c_\Phi}{\rho - \beta}, \qquad a_\xi := \frac{2L_A^2}{\rho^2}. \tag{59}$$

Assume explicitly that

$$\rho \in [\rho_-, \rho_+], \qquad \|y_k - y_\star\|_2 < r_{\mathrm{trg}}, \qquad \mathrm{dist}_F(P_k, \mathfrak{P}_\star) < r_{\mathrm{trg}}, \tag{60}$$

$$y_k, \widehat{y}_{k+1} \in \mathcal{Y}, \qquad \|\widehat{y}_{k+1} - y_\star\|_2 \le r_F, \qquad 0 < \tau \le 1. \tag{61}$$

Then the actual dual update satisfies

$$\begin{aligned} d_{k+1} \le & \left(1 - \frac{\mu_y}{2(\rho + \mu_y)} \tau\right) d_k \\ & + \tau \left(1 + \frac{\mu_y}{2\rho}\right) \frac{2c_\Phi}{\rho + \mu_y} \sigma_k \\ & + \tau \left(1 + \frac{2\rho}{\mu_y}\right) \frac{L_A^2}{\rho^2} \xi_k, \end{aligned} \tag{62}$$

and

$$\|y_{k+1} - y_k\|_2^2 \le \tau^2 (a_d d_k + a_\sigma \sigma_k + a_\xi \xi_k). \tag{63}$$

Define the pre-refresh spectral error and the one-V gains

$$\widetilde{\sigma}_{k+1} := \Phi_s(H(y_{k+1}), V_k), \qquad g_V := C_0 + \frac{C_1^2}{\kappa_V \eta_V}, \qquad q_V := 1 - \kappa_V \eta_V. \tag{64}$$

Lemma 5.16 (Moving-H and one-V recurrence)**.** *If $y_k, y_{k+1} \in \mathcal{Y}$ and $\eta_V > 0$, then*

$$\widetilde{\sigma}_{k+1} \le \left(1 + \frac{\kappa_V \eta_V}{4}\right) \sigma_k + g_V \|y_{k+1} - y_k\|_2^2. \tag{65}$$

If, in addition,

$$\widetilde{\sigma}_{k+1} \le \frac{\Delta^2}{2\overline{\Gamma}}, \qquad 0 < \eta_V \le \overline{\Gamma}^{-1}, \tag{66}$$

and if all premises (60)– (61) of Lemma 5.15 hold, then

$$\begin{aligned} \sigma_{k+1} \le q_V \Big[& \left(1 + \frac{\kappa_V \eta_V}{4}\right) \sigma_k \\ & + g_V \tau^2 (a_d d_k + a_\sigma \sigma_k + a_\xi \xi_k) \Big]. \end{aligned} \tag{67}$$

For the third recursion, write

$$\begin{aligned} X_{k+1}^{\sharp} &:= X_{\rho}^{\sharp}(y_{k+1}, P_{k+1}), \\ R_k &:= P_{k+1} X_k^{\sharp} P_{k+1} - X_{k+1}^{\sharp}. \end{aligned} \tag{68}$$

Then

$$\widehat{X}_{k+1} - X_{k+1}^{\sharp} = P_{k+1}(X_k - X_k^{\sharp})P_{k+1} + R_k. \tag{69}$$

Define

$$q_W := \frac{1 + \mu_X \eta_X/2}{1 + \mu_X \eta_X}, \qquad h_W := \frac{1 + 2/(\mu_X \eta_X)}{1 + \mu_X \eta_X}, \tag{70}$$

and

$$\begin{aligned} r_d &:= (2L_{X,1}^2 + 4C_T\eta_V^2)\tau^2 a_d + 4C_T\eta_V^2 + 2C_T\eta_V^2 g_V \tau^2 a_d, \\ r_\sigma &:= (2L_{X,1}^2 + 4C_T\eta_V^2)\tau^2 a_\sigma + 2C_T\eta_V^2 \left(1 + \frac{\kappa_V \eta_V}{4} + g_V \tau^2 a_\sigma\right), \\ r_\xi &:= (2L_{X,1}^2 + 4C_T\eta_V^2)\tau^2 a_\xi + 2C_T\eta_V^2 g_V \tau^2 a_\xi. \end{aligned} \tag{71}$$

Lemma 5.17 (Moving-target and one-X recurrence)**.** *Assume the hypotheses of Lemmas 5.15 and 5.16, including the second-part hypotheses of the latter. In addition, assume explicitly that*

$$\rho \in [\rho_-, \rho_+], \tag{72}$$

$$\|y_k - y_\star\|_2 < r_{\mathrm{trg}}, \qquad \|y_{k+1} - y_\star\|_2 < r_{\mathrm{trg}}, \qquad \mathrm{dist}_F(P_k, \mathfrak{P}_\star) < r_{\mathrm{trg}}, \tag{73}$$

$$\|y_{k+1} - y_\star\|_2 \le r_{\mathrm{drift}}, \qquad \widetilde{\sigma}_{k+1} \le r_{\mathrm{drift}}^2, \qquad 0 < \eta_V \le \bar{\eta}_{\mathrm{drift}}, \tag{74}$$

and

$$\mathrm{dist}_F(P_{k+1}, \mathfrak{P}_\star) < r_{\mathrm{trg}}, \qquad 0 < \eta_X \le L_A^{-2}. \tag{75}$$

Then

$$\xi_{k+1} \le h_W r_d d_k + h_W r_\sigma \sigma_k + (q_W + h_W r_\xi)\xi_k. \tag{76}$$

The three recurrences expose the coupling: the dual step is perturbed by model and primal tracking errors, the subspace step tracks the matrix moved by the dual update, and the primal step tracks the reduced target moved by both updates. In schematic form,

$$\begin{bmatrix} d_{k+1} \\ \sigma_{k+1} \\ \xi_{k+1} \end{bmatrix} \lesssim \underbrace{\begin{bmatrix} \text{proximal contraction} & \text{model error} & \text{inexact } X \\ \text{moving } H & \text{Riemannian contraction} & \text{dual-step error} \\ \text{target drift} & \text{target drift} & \text{PG contraction} \end{bmatrix}}_{\text{nonnegative coupling matrix}} \begin{bmatrix} d_k \\ \sigma_k \\ \xi_k \end{bmatrix}.$$

The diagonal entries are the three contraction mechanisms, and the off-diagonal entries measure target drift and inexact tracking.

Positive comparison system and localization. Fix the actual parameters for the entire run and impose

$$0 < \tau \le 1, \qquad 0 < \eta_V \le \min\{\overline{\Gamma}^{-1}, \bar{\eta}_{\mathrm{drift}}\}, \qquad 0 < \eta_X \le L_A^{-2}. \tag{77}$$

Define $M = M(\rho, \tau, \eta_V, \eta_X) \in \mathbb{R}_+^{3\times 3}$ by

$$\begin{aligned} M_{11} &= 1 - \frac{\mu_y}{2(\rho+\mu_y)}\tau, \\ M_{12} &= \tau\left(1+\frac{\mu_y}{2\rho}\right)\frac{2c_\Phi}{\rho+\mu_y}, \\ M_{13} &= \tau\left(1+\frac{2\rho}{\mu_y}\right)\frac{L_A^2}{\rho^2}, \end{aligned} \tag{78}$$

$$\begin{aligned} M_{21} &= q_V g_V \tau^2 a_d, \\ M_{22} &= q_V\left(1+\frac{\kappa_V\eta_V}{4}+g_V\tau^2 a_\sigma\right), \\ M_{23} &= q_V g_V \tau^2 a_\xi, \end{aligned} \tag{79}$$

and

$$M_{31} = h_W r_d, \qquad M_{32} = h_W r_\sigma, \qquad M_{33} = q_W + h_W r_\xi. \tag{80}$$

All quantities in (78)–(80) are evaluated at the actual fixed parameter values. The matrix is entrywise nonnegative: in particular, $0 < \kappa_V\eta_V \le 1/5$ and hence $q_V \ge 4/5$.

We give an explicit localization radius that verifies every local premise before the corresponding local estimate is invoked. Set

$$B_{\Delta y} := a_d + a_\sigma + a_\xi, \qquad B_{\mathrm{pre}} := 1 + \frac{\kappa_V\eta_V}{4} + g_V\tau^2 B_{\Delta y}, \tag{81}$$

and let

$$S_1 := M_{11} + M_{12} + M_{13}, \qquad S_2 := M_{21} + M_{22} + M_{23}.$$

Define

$$\begin{aligned} R_0 := \frac{1}{2}\min\Bigg\{ & \frac{r_{\mathrm{track}}^2}{4}, \frac{r_{\mathrm{trg}}^2}{4C_P^2}, r_{\mathrm{trg}}^2, R_Y^2, \frac{\min\{R_Y, r_F\}^2}{4}, \\ & \frac{\min\{R_Y, r_{\mathrm{trg}}, r_{\mathrm{drift}}\}^2}{(1+\tau\sqrt{B_{\Delta y}})^2}, \\ & \frac{\min\{\underline{\Delta}^2/(2\overline{\Gamma}), r_{\mathrm{drift}}^2\}}{B_{\mathrm{pre}}}, \\ & \frac{\min\{r_{\mathrm{track}}, r_{\mathrm{trg}}/C_P\}^2}{(\sqrt{S_1}+\sqrt{S_2})^2}\Bigg\} > 0. \end{aligned} \tag{82}$$

Theorem 5.18 (Localized comparison inequality)**.** *Suppose Assumptions 3.3 and 5.7 hold, (52) and (77) hold, and*

$$d_k \le R_0, \qquad \sigma_k \le R_0. \tag{83}$$

Then the current target $X_k^\sharp = X_\rho^\sharp(y_k, P_k)$ exists uniquely. Define $\xi_k = \left\|X_k - X_k^\sharp\right\|_F^2$ and suppose in addition that

$$\xi_k \le R_0. \tag{84}$$

All targets and all local estimates needed for the update from k to $k+1$ are then well defined. In particular, the next reduced target $X_{k+1}^\sharp = X_\rho^\sharp(y_{k+1}, P_{k+1})$ exists uniquely, so all three next-iterate errors d_{k+1}, σ_{k+1}, and $\xi_{k+1} = \left\|X_{k+1} - X_{k+1}^\sharp\right\|_F^2$ are well defined, and, componentwise,

$$\begin{bmatrix} d_{k+1} \\ \sigma_{k+1} \\ \xi_{k+1} \end{bmatrix} \le M \begin{bmatrix} d_k \\ \sigma_k \\ \xi_k \end{bmatrix}. \tag{85}$$

Theorem 5.18 chooses one common radius that validates every local hypothesis before it is used. Within this radius, the three scalar recurrences become the positive comparison inequality (85) without circular locality assumptions.

Stability and convergence.

Theorem 5.19 (Feasible fixed parameters)**.** *Fix any actual $\rho \in [\rho_-, \rho_+]$ and $\eta_X \in (0, L_A^{-2}]$. Set*

$$\eta_V = \varepsilon, \qquad \tau = \varepsilon^2, \qquad w_\varepsilon = (1, \sqrt{\varepsilon}, \sqrt{\varepsilon})^\top. \tag{86}$$

There exists $\varepsilon_0 \in (0, 1]$ such that, for every $0 < \varepsilon \le \varepsilon_0$, the basic step bounds (77) hold and

$$w_\varepsilon^\top M \le q_\varepsilon w_\varepsilon^\top, \qquad q_\varepsilon := 1 - \frac{\mu_y}{4(\rho + \mu_y)}\varepsilon^2 \in (0, 1). \tag{87}$$

The scaling in Theorem 5.19 separates the contraction and coupling scales. The explicit left weight absorbs the coupling terms while allowing a fixed one-X step.

Theorem 5.20 (Fixed-work local linear convergence)**.** *Suppose Assumptions 3.3 and 5.7 hold, the proximal interval satisfies (52), and the actual fixed parameters satisfy (77). Let $w \in \mathbb{R}^3_{++}$ and $q \in (0, 1)$ satisfy*

$$w^\top M \le q w^\top \tag{88}$$

and write $w_{\min} := \min\{w_1, w_2, w_3\} > 0$. Assume

$$\sqrt{d_0} + \sqrt{\sigma_0} \le r_{\text{track}}, \tag{89}$$

so that $X_0^\sharp = X_\rho^\sharp(y_0, P_0)$ exists uniquely by Theorem E.31. Suppose

$$\mathcal{L}_0 = w_1 d_0 + w_2 \sigma_0 + w_3 \xi_0 \le R_0 w_{\min}. \tag{90}$$

Then, for every $k \ge 0$, the exact reduced target $X_k^\sharp = X_\rho^\sharp(y_k, P_k)$ exists uniquely. Define

$$\mathcal{L}_k := w_1 d_k + w_2 \sigma_k + w_3 \xi_k. \tag{91}$$

Then,

$$\mathcal{L}_k \le q^k \mathcal{L}_0. \tag{92}$$

Consequently,

$$\|y_k - y_\star\|_2 = O(q^{k/2}), \qquad \|X_k - X_\star\|_F = O(q^{k/2}). \tag{93}$$

Theorem 5.20 closes the comparison argument in an invariant sublevel set of the weighted scalar error. It gives local linear convergence with one dual update, one residual/QR update, and one reduced projected-gradient update per outer iteration.

Appendix

A Proofs for Section 3

This section contains the proofs of the results stated in Section 3, in their order of appearance.

Proof of Lemma 3.2. Let $G = Q \operatorname{diag}(\lambda_i) Q^\top$ and set $\widetilde{X} = Q^\top X Q \succeq 0$. Then

$$\langle G, X \rangle = \sum_i \lambda_i \widetilde{X}_{ii} \leq \max\{0, \lambda_1(G)\} \sum_i \widetilde{X}_{ii} \leq \max\{0, \lambda_1(G)\}.$$

The stated choices attain the bound. Equality in both inequalities gives the characterization of maximizers. □

Proof of Lemma 3.5. If $W \in \mathbb{S}^r$, then $P(VWV^\top)P = VWV^\top$, so $VWV^\top \in \mathcal{E}(P)$. Conversely, if $X \in \mathcal{E}(P)$, then $X = PXP = V(V^\top X V)V^\top$ with $V^\top X V \in \mathbb{S}^r$. This proves the identity for $\mathcal{E}(P)$. Since $V^\top V = I_r$, $W \succeq 0$ if and only if $VWV^\top \succeq 0$, and $\operatorname{tr}(VWV^\top) = \operatorname{tr} W$. Applying these facts to the two representations proves the identity for $\mathcal{D}(P)$. □

Proof of Proposition 3.7. The set $\mathcal{D}(P)$ is nonempty, compact, and convex, so the continuous convex objective $\Psi_{\rho,y}$ has a minimizer X^+. Put $g := \mathcal{A}(X^+) - b$ and $\widetilde{z} := y - g/\rho$. Its first-order optimality condition is

$$\left\langle \nabla \Psi_{\rho,y}(X^+), X - X^+ \right\rangle \geq 0 \qquad \text{for every } X \in \mathcal{D}(P).$$

Since $\nabla \Psi_{\rho,y}(X^+) = \mathcal{A}^* g - \rho H(y) = -\rho H(\widetilde{z})$, this condition says that X^+ maximizes $\langle H(\widetilde{z}), X \rangle$ over $\mathcal{D}(P)$. By Lemma 3.5, it therefore attains the maximum defining $M_V(\widetilde{z})$. For every $z \in \mathbb{R}^m$, the same feasible matrix gives

$$M_V(z) \geq -b^\top z + \left\langle H(z), X^+ \right\rangle = M_V(\widetilde{z}) + \langle g, z - \widetilde{z} \rangle .$$

Thus $g \in \partial M_V(\widetilde{z})$ and $0 \in \partial M_V(\widetilde{z}) + \rho(\widetilde{z} - y)$. The model-proximal objective is ρ-strongly convex, so $\widetilde{z} = z^+$ is its unique minimizer. This proves (13) for every reduced minimizer X^+. □

B Proofs for Section 4

This section contains the proofs of the results stated in Section 4, in their order of appearance.

Proof of Lemma 4.1. Compression preserves positive semidefiniteness, and

$$\operatorname{tr}(PXP) = \operatorname{tr}(X^{1/2} P X^{1/2}) \leq \operatorname{tr} X$$

because $0 \preceq P \preceq I$. Hence the compression is feasible. For any $Y \in \mathcal{E}(P)$,

$$X - Y = (PXP - Y) + (X - PXP),$$

and the two terms are Frobenius-orthogonal in the block decomposition $\operatorname{Range}(P) \oplus \operatorname{Null}(P)$. Pythagoras proves that the feasible compression is the unique nearest point even in the smaller set $\mathcal{D}(P)$. □

Proof of Corollary 4.2. For every $Y \in \mathcal{E}(P)$, the block decomposition associated with $\mathrm{Range}(P) \oplus \mathrm{Null}(P)$ gives the orthogonal identity

$$\|X - Y\|_F^2 = \|PXP - Y\|_F^2 + \|X - PXP\|_F^2 .$$

The second term is independent of Y. By Lemma 3.5, write $Y = VWV^\top$ and use the isometry $W \mapsto VWV^\top$ to obtain

$$\operatorname*{argmin}_{Y \in \mathcal{D}(P)} \|X - Y\|_F^2 = V \left(\operatorname*{argmin}_{W \in \Theta_r} \left\| V^\top X V - W \right\|_F^2 \right) V^\top .$$

This proves (18). The set Θ_r is orthogonally invariant, so the nearest matrix commutes with $V^\top XV$; reducing to its eigenvalues gives (19). □

Proof of Lemma 4.4. Since $V^\top D = 0$, $Z^\top Z = I + \eta^2 D^\top D \succ 0$; hence Z has full column rank and $Z(Z^\top Z)^{-1}Z^\top$ is its range projector. If $V' = VO$ with $O \in O(r)$, then $D' = DO$ and $Z' = ZO$, so the projector is unchanged. The Ritz rotation in the algorithm only changes the basis of the same space, and thin QR returns exactly the range projector in (21). □

C Proofs for Section 5.1

This section isolates the mechanism that contracts the history-subspace error. It first fixes one spectral neighborhood and one set of constants that will be used everywhere in the paper. It then introduces the three fixed-matrix quantities Φ_s, e_s, and $\mathcal{R}_s$, proves their precise relations, controls the change of Φ_s when $H(y)$ moves, and finally establishes the one-full-residual contraction. No iterate-dependent recurrence is derived in this section; coupling with the dual and primal errors is postponed to Section 5.4.

C.1 Strict complementarity and one uniform spectral neighborhood

Proof of Lemma 5.1. From $S_\star X_\star = 0$ and the invertibility of $\Sigma_\star$, $S_\star U_\star = 0$, so $\mathrm{Range}(U_\star) \subseteq \mathrm{Null}(S_\star)$. Strict complementarity gives $\mathrm{rank}\, S_\star = n - s$, and both subspaces therefore have dimension s; they coincide. The restriction of $S_\star \succeq 0$ to $\mathrm{Range}(U_\star)^\perp$ is positive definite. Let $\delta_\star$ be its least eigenvalue. This proves (24). Since (S1) gives $H(y_\star) = -S_\star$, negating the two diagonal blocks yields (25). □

Proof of Lemma 5.2. For $y \in \mathcal{Y}$, the affine form of H and (27) give

$$\|H(y) - H(y_\star)\|_2 \le L_H \|y - y_\star\|_2 \le \frac{\Delta_\star}{16}. \tag{94}$$

Applying Weyl's inequality separately to the sth and $(s+1)$st eigenvalues yields

$$\begin{aligned} \lambda_s(H(y)) - \lambda_{s+1}(H(y)) &\ge \Delta_\star - 2\|H(y) - H(y_\star)\|_2 \\ &\ge \Delta_\star - \frac{\Delta_\star}{8} = \underline{\Delta}. \end{aligned}$$

The spectral width is continuous, and $\mathcal{Y}$ is compact, so the maximum in (29) is finite.

For $y, z \in \mathcal{Y}$,

$$\|H(z) - H(y)\|_2 \le L_H \|z - y\|_2 \le 2L_H R_Y \le \frac{\Delta_\star}{8} < \frac{\underline{\Delta}}{4}. \tag{95}$$

The leading s-dimensional invariant subspaces are therefore separated from the remaining spectrum along the entire segment between $H(y)$ and $H(z)$. The standard Davis–Kahan projector estimate [11, 32] gives

$$\|P_s(H(y)) - P_s(H(z))\|_F \le \frac{2\sqrt{2s}}{\underline{\Delta}} \|H(y) - H(z)\|_2 \le L_P \|y - z\|_2 .$$

□

C.2 Hidden Ky Fan gap, angle error, and active residual

Proof of Proposition 5.4. If $Q \in \mathrm{St}(n, s)$ and $\mathrm{Range}(Q) \subseteq \mathrm{Range}(V)$, then $Q = VY$ with $Y = V^\top Q$ and $Y^\top Y = I_s$. Conversely, $Y \in \mathrm{St}(r, s)$ gives $VY \in \mathrm{St}(n, s)$ with range contained in $\mathrm{Range}(V)$. Thus $Y \mapsto VY$ is a bijection onto the feasible set defining $Q_s(H, V)$, and

$$\mathrm{tr}((VY)^\top H(VY)) = \mathrm{tr}(Y^\top V^\top HVY) = \mathrm{tr}(Y^\top BY).$$

Ky Fan's maximum principle now gives the stated maximizers and the formula for $\Phi_s(H, V)$. □

Lemma C.1 (The hidden Ritz gap and subspace angle are equivalent)**.** *Let $H \in \mathbb{S}^n$ and $V \in \mathrm{St}(n, r)$. Suppose $\Delta(H) > 0$ and put $\Gamma(H) := \lambda_1(H) - \lambda_n(H) > 0$. Then*

$$\Delta(H) e_s(H, V) \le \Phi_s(H, V) \le \Gamma(H) e_s(H, V). \tag{96}$$

Moreover,

$$\left\| (I - VV^\top) U_s(H) \right\|_F^2 \le \frac{\Phi_s(H, V)}{\Delta(H)}. \tag{97}$$

Proof. Let u_i be an orthonormal eigenbasis of H and set $a_i = u_i^\top QQ^\top u_i$, where $Q = Q_s(H, V)$. Since $QQ^\top$ is a rank-s projector, $0 \le a_i \le 1$ and

$$\sum_{i=1}^n a_i = \mathrm{tr}(QQ^\top) = s. \tag{98}$$

Moreover,

$$e_s(H, V) = \mathrm{tr}\big(Q^\top (I - P_s(H))Q\big) = \sum_{j>s} a_j, \tag{99}$$

$$\sum_{i \le s} (1 - a_i) = s - \sum_{i \le s} a_i = \sum_{j>s} a_j = e_s(H, V). \tag{100}$$

Expanding the two Ky Fan traces in the eigenbasis of H gives

$$\Phi_s(H, V) = \sum_{i \le s} \lambda_i(H)(1 - a_i) - \sum_{j>s} \lambda_j(H) a_j. \tag{101}$$

Use $\lambda_i \ge \lambda_s$ for $i \le s$ and $\lambda_j \le \lambda_{s+1}$ for $j > s$. Equations (100)–(101) yield

$$\Phi_s(H, V) \ge [\lambda_s(H) - \lambda_{s+1}(H)] e_s(H, V) = \Delta(H) e_s(H, V).$$

The reverse estimates $\lambda_i \le \lambda_1$ and $\lambda_j \ge \lambda_n$ similarly give $\Phi_s(H, V) \le \Gamma(H) e_s(H, V)$.

It remains to relate the full history space to the best restricted s-space. Because $\mathrm{Range}(Q) \subseteq \mathrm{Range}(V)$, $QQ^\top \preceq VV^\top$, and hence

$$\left\|(I - VV^\top)U_s\right\|_F^2 \leq \left\|(I - QQ^\top)U_s\right\|_F^2 = e_s.$$

The last equality is the equal-rank projector identity applied to $P_s(H) = U_s U_s^\top$ and $QQ^\top$. Combining this estimate with the lower gap–angle bound proves (97). □

Lemma C.2 (Hidden active-residual identity)**.** *Let $H \in \mathbb{S}^n$, $V \in \mathrm{St}(n, r)$, and $Q = Q_s(H, V)$. Then*

$$(I - VV^\top)HQ = (I - QQ^\top)HQ. \tag{102}$$

Proof. Put $Y := V^\top Q \in \mathrm{St}(r, s)$, so $Q = VY$. The range of a Ky Fan maximizing Y is an invariant subspace of $B := V^\top HV$, even when its columns are not individual eigenvectors. Consequently,

$$BY = Y(Y^\top BY) = Y(Q^\top HQ).$$

It follows that

$$VV^\top HQ = VBY = Q(Q^\top HQ) = QQ^\top HQ.$$

Subtract this equality from HQ to obtain (102). □

Lemma C.3 (Residual energy controls the hidden Ritz gap)**.** *Let $H \in \mathbb{S}^n$ and $V \in \mathrm{St}(n, r)$ satisfy $\Delta(H) > 0$ and $\Gamma(H) > 0$. If*

$$\Phi_s(H, V) \leq \frac{\Delta(H)^2}{2\Gamma(H)}, \tag{103}$$

then

$$\|\mathcal{R}_s(H, V)\|_F^2 \geq \frac{\Delta(H)^2}{4\Gamma(H)} \Phi_s(H, V). \tag{104}$$

Proof. Let $Q = Q_s(H, V)$, $B_Q = Q^\top HQ$, and write an eigenbasis of H as $[U, U_\perp]$ with U spanning its leading s-dimensional invariant subspace. Put $T := U_\perp^\top Q$. For every unit a and $x = Qa$, the trailing spectral mass of x satisfies

$$\left\|U_\perp^\top x\right\|_2^2 = \|Ta\|_2^2 \leq \|T\|_F^2 = e_s(H, V).$$

Decomposing x into its leading and trailing components therefore gives

$$\begin{aligned} a^\top B_Q a = x^\top H x &\geq \lambda_s(H) \left\|U^\top x\right\|_2^2 + \lambda_n(H) \left\|U_\perp^\top x\right\|_2^2 \\ &\geq \lambda_s(H) - \Gamma(H) e_s(H, V). \end{aligned} \tag{105}$$

Lemma C.1 and (103) imply $e_s \leq \Delta/(2\Gamma)$, hence

$$\lambda_{\min}(B_Q) \geq \lambda_s(H) - \Delta(H)/2. \tag{106}$$

Let $\Lambda_2 := U_\perp^\top H U_\perp$. Using Lemma C.2 and $Q^\top HQ = B_Q$ gives the Sylvester identity

$$U_\perp^\top \mathcal{R}_s = \Lambda_2 T - T B_Q. \tag{107}$$

Every eigenvalue of Λ_2 is at most $\lambda_{s+1}(H) = \lambda_s(H) - \Delta(H)$, whereas every eigenvalue of B_Q is at least $\lambda_s(H) - \Delta(H)/2$. Orthogonal diagonalization makes the quantitative implication explicit. If $O_2^\top \Lambda_2 O_2 = \mathrm{diag}(\alpha_i)$, $O_1^\top B_Q O_1 = \mathrm{diag}(\beta_j)$, and $\widehat{T} = O_2^\top T O_1$, then

$$\|\Lambda_2 T - T B_Q\|_F^2 = \sum_{i,j} (\alpha_i - \beta_j)^2 \widehat{T}_{ij}^2 \geq \frac{\Delta(H)^2}{4} \|T\|_F^2 .$$

Since orthogonal projection cannot increase the Frobenius norm,

$$\begin{aligned} \|\mathcal{R}_s\|_F^2 &\geq \left\| U_\perp^\top \mathcal{R}_s \right\|_F^2 \geq \frac{\Delta(H)^2}{4} \|T\|_F^2 \\ &= \frac{\Delta(H)^2}{4} e_s(H, V) \geq \frac{\Delta(H)^2}{4\Gamma(H)} \Phi_s(H, V), \end{aligned}$$

where the last inequality is the upper gap–angle estimate. □

C.3 Motion of the spectral matrix and one residual step

Proof of Lemma 5.5. Let $P_y = P_s(H(y))$, $P_z = P_s(H(z))$, and $Q = Q_s(H(y), V)$. Ky Fan optimality gives

$$\Phi_s(H(z), V) \leq \Phi_s(H(y), V) + \left\langle H(z) - H(y), P_z - QQ^\top \right\rangle . \tag{108}$$

For completeness, (108) follows by using Q as a feasible competitor for the restricted Ky Fan problem at $H(z)$ and P_y as the global maximizer at $H(y)$:

$$\begin{aligned} \Phi_s(H(z), V) &\leq \left\langle H(z), P_z - QQ^\top \right\rangle \\ &= \left\langle H(y), P_z - QQ^\top \right\rangle + \left\langle H(z) - H(y), P_z - QQ^\top \right\rangle \\ &\leq \left\langle H(y), P_y - QQ^\top \right\rangle + \left\langle H(z) - H(y), P_z - QQ^\top \right\rangle . \end{aligned}$$

Now split $P_z - QQ^\top = (P_z - P_y) + (P_y - QQ^\top)$. For the first term, spectral–nuclear norm duality, the rank bound $\mathrm{rank}(P_z - P_y) \leq 2s$, and the projector Lipschitz estimate imply

$$\begin{aligned} |\langle H(z) - H(y), P_z - P_y \rangle| &\leq \|H(z) - H(y)\|_2 \, \|P_z - P_y\|_* \\ &\leq L_H \, \|z - y\|_2 \, \sqrt{2s} \, \|P_z - P_y\|_F \\ &\leq C_0 \, \|z - y\|_2^2 . \end{aligned} \tag{109}$$

For two rank-s projectors, the equal-rank identities give

$$\left\| P_y - QQ^\top \right\|_F = \sqrt{2} \, \|(I - P_y) Q\|_F , \qquad \mathrm{rank}(P_y - QQ^\top) \leq 2s.$$

Consequently,

$$\left\| P_y - QQ^\top \right\|_* \leq 2\sqrt{s} \, \|(I - P_y) Q\|_F \leq 2 \sqrt{\frac{s}{\Delta}} \sqrt{\Phi_s(H(y), V)}. \tag{110}$$

Use again $\|H(z) - H(y)\|_2 \leq L_H \|z - y\|_2$, then substitute (109) and (110) into (108). The definitions of C_0 and C_1 give (33). □

Lemma C.4 (Quantitative Ritz ascent for any fixed small step)**.** *Let $H \in \mathbb{S}^n$, $V \in \mathrm{St}(n,r)$, $\eta > 0$, $Q = Q_s(H,V)$, $R = \mathcal{R}_s(H,V)$, and*

$$Z_\eta = (Q + \eta R)(I + \eta^2 R^\top R)^{-1/2}.$$

If $0 < \eta\Gamma(H) \le 1$, then

$$\mathrm{tr}(Z_\eta^\top H Z_\eta) - \mathrm{tr}(Q^\top H Q) \ge \frac{4}{5}\eta \left\|R\right\|_F^2. \tag{111}$$

Proof. The update is orthonormal because $Q^\top R = 0$. A scalar shift of H changes neither the Rayleigh-trace difference nor R. Shift by the midpoint of the extreme eigenvalues, so that $\left\|H\right\|_2 = M := \Gamma(H)/2$. The Ritz relation has the form

$$HQ = Q\Lambda + R, \qquad Q^\top H R = R^\top H Q = R^\top R =: B.$$

With $C_\eta = (I + \eta^2 B)^{-1}$, direct expansion gives

$$\begin{aligned}\mathrm{tr}(Z_\eta^\top H Z_\eta) - \mathrm{tr}(Q^\top H Q) = 2\eta\,\mathrm{tr}(C_\eta B) + \eta^2\,\mathrm{tr}(C_\eta R^\top H R)\\ - \eta^2\,\mathrm{tr}(C_\eta B\Lambda).\end{aligned}$$

Since $\left\|H\right\|_2, \left\|\Lambda\right\|_2 \le M$, the last two terms are together at least $-2M\eta^2\,\mathrm{tr}(C_\eta B)$. The step condition gives $2\eta - 2M\eta^2 \ge \eta$. Moreover $\left\|R\right\|_2 \le M$, so $\eta^2\left\|B\right\|_2 \le 1/4$ and therefore $C_\eta \succeq (4/5)I$. Hence the increase is at least $(4/5)\eta\,\mathrm{tr}\,B = (4/5)\eta\left\|R\right\|_F^2$. □

Proof of Theorem 5.6. After the descending Ritz rotation, Q consists of the first s columns and $\mathcal{R}_s$ consists of the first s residual columns. More precisely, if $\widetilde{V} = VY = [Q, Q_\perp]$, then

$$(I - \widetilde{V}\widetilde{V}^\top)H\widetilde{V} = [\mathcal{R}_s, R_\perp].$$

The QR update does not change the range of its input, and therefore the new history space contains the normalized block

$$Z_s = (Q + \eta_V \mathcal{R}_s)(I + \eta_V^2 \mathcal{R}_s^\top \mathcal{R}_s)^{-1/2}.$$

Let $\mathrm{KF}_s(H;\mathcal{V})$ denote the maximal Ky Fan trace over s-dimensional subspaces contained in $\mathcal{V}$. Since $\mathrm{Range}(Z_s) \subseteq \mathrm{Range}(V^+)$,

$$\begin{aligned}\Phi_s(H, V^+) &\le \sum_{i=1}^{s} \lambda_i(H) - \mathrm{tr}(Z_s^\top H Z_s)\\ &= \Phi_s(H,V) - \left[\mathrm{tr}(Z_s^\top H Z_s) - \mathrm{tr}(Q^\top H Q)\right].\end{aligned}$$

By Lemma C.4, the bracket is at least $(4/5)\eta_V \left\|\mathcal{R}_s\right\|_F^2$. Under the stated gap threshold, Lemma C.3 gives

$$\frac{4}{5}\eta_V \left\|\mathcal{R}_s\right\|_F^2 \ge \eta_V \frac{\Delta(H)^2}{5\Gamma(H)}\Phi_s(H,V).$$

Substitution proves (34). On $\mathcal{Y}$, $\Delta(H) \ge \underline{\Delta}$ and $\Gamma(H) \le \overline{\Gamma}$; moreover $\eta_V\overline{\Gamma} \le 1$. These inequalities give the uniform factor $1 - \kappa_V\eta_V$. □

D Proofs for Section 5.2

The second contraction mechanism concerns the dual variable. Two model estimates are needed. The first is global and controls the displacement of an exact model-proximal point. The second is local and converts the hidden Ritz gap at the model center into a quadratic model error at a nearby dual point. They will be combined with dual quadratic growth to contract the exact model-proximal point in Section 5.4. The actual iterate uses the inexact target X_k; its perturbation is treated there as well.

D.1 Global model Lipschitz and displacement bounds

Proposition D.1 (Affine Danskin formula)**.** *For every $V \in \mathrm{St}(n,r)$ and $z \in \mathbb{R}^m$,*

$$\partial M_V(z) = \{\mathcal{A}(VWV^\top) - b : W \in \operatorname*{argmax}_{W \in \Theta_r} \left\langle V^\top H(z) V, W \right\rangle\}.$$

Justification of Proposition D.1. Equation (9) is a maximum of affine functions indexed by the compact set Θ_r. The affine form of Danskin's theorem [10] therefore identifies its subdifferential with the convex hull of the active affine slopes. The active coefficient set is convex, and $W \mapsto \mathcal{A}(VWV^\top) - b$ is affine, so that image is already convex. This gives the displayed formula in Proposition D.1. □

Proof of Lemma 5.8. Write the model as an affine maximum,

$$M_V(z) = \max_{W \in \Theta_r} \left\{ \left\langle \mathcal{A}(VWV^\top) - b, z \right\rangle - \left\langle C, VWV^\top \right\rangle \right\}. \tag{112}$$

By Proposition D.1, every $g \in \partial M_V(z)$ has the form

$$g = \mathcal{A}(VWV^\top) - b \tag{113}$$

for some maximizing $W \in \Theta_r$. Since $W \succeq 0$,

$$\|W\|_F = \left(\sum_i \lambda_i(W)^2 \right)^{1/2} \le \sum_i \lambda_i(W) = \operatorname{tr} W \le 1.$$

The identity $\left\|VWV^\top\right\|_F = \|W\|_F$, which follows from $V^\top V = I_r$, and the definition of L_A therefore give

$$\|g\|_2 \le L_A \left\|VWV^\top\right\|_F + \|b\|_2 = L_A \|W\|_F + \|b\|_2 \le G_M. \tag{114}$$

Choose $g \in \partial M_V(z)$ and $g' \in \partial M_V(z')$. The two subgradient inequalities imply

$$\left\langle g', z - z' \right\rangle \le M_V(z) - M_V(z') \le \left\langle g, z - z' \right\rangle.$$

Applying Cauchy–Schwarz and (114) on both sides proves the global Lipschitz bound.

For the proximal displacement, optimality of $z_\rho = z_\rho(y, V)$ gives

$$0 \in \partial M_V(z_\rho) + \rho(z_\rho - y).$$

Hence there exists $g_\rho \in \partial M_V(z_\rho)$ with $g_\rho + \rho(z_\rho - y) = 0$. Taking norms and using (114) yields

$$\rho \|z_\rho - y\|_2 = \|g_\rho\|_2 \le G_M,$$

which is the second estimate in (39). □

D.2 From the hidden Ritz gap to local model accuracy

Proof of Lemma 5.9. Let $\lambda_1 \geq \cdots \geq \lambda_n$ and $\mu_1 \geq \cdots \geq \mu_r$ be the eigenvalues of $H(z)$ and $V^\top H(z)V$, respectively. Rayleigh–Ritz gives $\mu_i \leq \lambda_i$ for $1 \leq i \leq r$. Writing $[a]_+ := \max\{a, 0\}$, Lemma 3.2 and Proposition 5.4 imply

$$\begin{aligned} 0 \leq F(z) - M_V(z) &= [\lambda_1]_+ - [\mu_1]_+ \\ &\leq \lambda_1 - \mu_1 \leq \sum_{i=1}^{s} (\lambda_i - \mu_i) = \Phi_s(H(z), V). \end{aligned} \tag{115}$$

Let $\delta := \|z - y\|_2$ and $\phi := \Phi_s(H(y), V)$. The moving-gap bound Lemma 5.5 and $C_1\delta\sqrt{\phi} \leq \phi + (C_1^2/4)\delta^2$ give

$$F(z) - M_V(z) \leq 2\phi + \left(C_0 + \frac{C_1^2}{4}\right)\delta^2. \tag{116}$$

By (29) and (32),

$$C_0 + \frac{C_1^2}{4} = \frac{5sL_H^2}{\underline{\Delta}} = \frac{\beta}{2}.$$

Thus (116) is exactly (41) with the constants in (40). □

E Proofs for Section 5.3

Throughout this section, Assumption 3.3 is in force. Thus (S1)–(S4) may be invoked without repeating the assumption in every statement.

Lemma E.1 (Slack tangent lineality and its active orthogonal complement)**.** *Let $S \in \mathbb{S}_+^n$ have rank $n - s$. Choose $U \in \mathrm{St}(n, s)$ and $N \in \mathrm{St}(n, n - s)$ so that $[U, N]$ is orthogonal and $S = N\Lambda N^\top$ with $\Lambda \succ 0$. Then*

$$\operatorname{lin} T_{\mathbb{S}_+^n}(S) = \left\{ [U, N] \begin{bmatrix} 0 & B \\ B^\top & C \end{bmatrix} [U, N]^\top : B \in \mathbb{R}^{s\times(n-s)},\ C \in \mathbb{S}^{n-s} \right\}, \tag{117}$$

and

$$\left(\operatorname{lin} T_{\mathbb{S}_+^n}(S)\right)^\perp = \{UAU^\top : A \in \mathbb{S}^s\}. \tag{118}$$

Proof. In the displayed eigenbasis, the standard tangent-cone formula for the PSD cone is

$$T_{\mathbb{S}_+^n}(S) = \left\{ [U, N] \begin{bmatrix} A & B \\ B^\top & C \end{bmatrix} [U, N]^\top : A \succeq 0 \right\}; \tag{119}$$

see, for example, [3]. A direction and its negative both belong to this cone exactly when $A = 0$, which proves (117). Frobenius orthogonality to every free off-diagonal block B and lower-right block C forces those two blocks of a matrix in the orthogonal complement to vanish. Only the upper-left symmetric block remains, yielding (118). □

Lemma E.2 (Dual nondegeneracy gives quantitative active-block injectivity)**.** *Under (S4), there exists $\gamma_\star > 0$ such that*

$$\left\| \mathcal{A}(U_\star A U_\star^\top) \right\|_2 \geq \gamma_\star \|A\|_F \qquad \text{for every } A \in \mathbb{S}^s. \tag{120}$$

Proof. Taking orthogonal complements in (S4), using $\mathrm{Range}(\mathcal{A}^*)^\perp = \ker \mathcal{A}$ and Lemma E.1 with $(S, U) = (S_\star, U_\star)$, gives

$$\ker \mathcal{A} \cap \{U_\star A U_\star^\top : A \in \mathbb{S}^s\} = \{0\}.$$

Thus the linear map $\mathcal{L}_\star : A \mapsto \mathcal{A}(U_\star A U_\star^\top)$ is injective. The function $A \mapsto \|\mathcal{L}_\star(A)\|_2$ is continuous and strictly positive on the compact unit sphere $\{A \in \mathbb{S}^s : \|A\|_F = 1\}$. Its minimum

$$\gamma_\star := \min_{\|A\|_F = 1} \left\| \mathcal{A}(U_\star A U_\star^\top) \right\|_2 \tag{121}$$

is therefore positive. Homogeneity proves (120). □

Lemma E.3 (Compactness of the redundant projector family)**.** *$\mathfrak{P}_\star$ is compact in $\mathbb{S}^n$.*

Proof. $\mathfrak{P}_r$ is the common zero set of the continuous equations $P^\top = P$, $P^2 = P$, and $\operatorname{tr} P = r$, hence it is closed. Every $P \in \mathfrak{P}_r$ satisfies $\|P\|_F^2 = \operatorname{tr} P = r$, so $\mathfrak{P}_r$ is bounded and therefore compact. The additional equation $PP_\star = P_\star$ defines a closed subset. □

Lemma E.4 (The base reduced target equals $X_\star$ on all redundant projectors)**.** *For every $\rho > 0$ and $P \in \mathfrak{P}_\star$,*

$$X_\rho^\sharp(y_\star, P) = X_\star, \tag{122}$$

and the minimizer is globally unique.

Proof. Because $P \in \mathfrak{P}_\star$, $PX_\star P = X_\star$, so $X_\star \in \mathcal{D}(P)$. For $X \in \mathcal{D}(P)$ and $E = X - X_\star$, primal feasibility and $H(y_\star) = -S_\star$ give

$$\Psi_{\rho,y_\star}(X) - \Psi_{\rho,y_\star}(X_\star) = \frac{1}{2}\|\mathcal{A}(E)\|_2^2 + \rho \langle S_\star, X \rangle . \tag{123}$$

Both terms are nonnegative. Equality implies $S_\star X = 0$, hence $\mathrm{Range}(X) \subseteq \mathrm{Null}(S_\star) = \mathrm{Range}(U_\star)$. Thus $X = U_\star M U_\star^\top$ for some $M \succeq 0$. The other equality $\mathcal{A}(X - X_\star) = 0$ and Lemma E.2 yield $M = \Sigma_\star$. Therefore $X = X_\star$. □

E.1 A smooth reduced KKT branch

Proposition E.5 (Reduced KKT system)**.** *Let $\rho > 0$, $y \in \mathbb{R}^m$, and $P \in \mathfrak{P}_r$. A matrix $X \in \mathcal{D}(P)$ minimizes (10) if and only if there exist $Z \in \mathcal{E}(P)$ and $\widehat{\nu} \geq 0$ such that*

$$\begin{aligned} &P\nabla\Psi_{\rho,y}(X)P + \widehat{\nu}P - Z = 0, \qquad X, Z \succeq 0, \qquad XZ = 0, \\ &\operatorname{tr} X \leq 1, \qquad \widehat{\nu}(\operatorname{tr} X - 1) = 0, \qquad PXP = X, \quad PZP = Z, \end{aligned} \tag{124}$$

where

$$\nabla\Psi_{\rho,y}(X) = \mathcal{A}^*(\mathcal{A}(X) - b) - \rho H(y). \tag{125}$$

Proof. The natural Euclidean space is $\mathcal{E}(P)$. If $P = VV^\top$, then $P/(2r) = V(I_r/(2r))V^\top$ is positive definite relative to this space and has trace $1/2$, so Slater's condition holds. The restricted gradients of $\Psi_{\rho,y}$ and $\operatorname{tr}(\cdot)$ are $P\nabla\Psi_{\rho,y}(X)P$ and P. The standard convex KKT theorem in $\mathcal{E}(P)$, together with self-duality of its PSD cone, gives (124); PSD complementarity is equivalent to $XZ = 0$. Conversely, the same KKT theorem makes these conditions sufficient. □

Fix a base pair $\rho_0 \in [\rho_-, \rho_+]$ and $\bar{P} \in \mathfrak{P}_\star$. Choose $R \in \mathbb{R}^{n \times (r-s)}$ such that

$$\bar{V} := [U_\star, R] \in \mathrm{St}(n, r), \qquad \bar{P} = P_\star + RR^\top. \tag{126}$$

Lemma E.6 (Equivalent representations and positivity of the base slack)**.** *Let $\rho_0 \in [\rho_-, \rho_+]$, $\bar{P} \in \mathfrak{P}_\star$, and R satisfy (126). At the base point the reduced slack is*

$$Z_0 := \rho_0 \bar{P} S_\star \bar{P} = R N_0 R^\top, \qquad N_0 := \rho_0 R^\top S_\star R \succeq \rho_- \delta_\star I. \tag{127}$$

In particular, $N_0 \succ 0$, and $(X_\star, 0, Z_0)$ is the base reduced KKT triple.

Proof. Original complementarity and $\Sigma_\star \succ 0$ give $S_\star U_\star = 0$. Expanding $\bar{P} = P_\star + RR^\top$ therefore yields

$$\bar{P} S_\star \bar{P} = RR^\top S_\star RR^\top = R(R^\top S_\star R)R^\top, \tag{128}$$

which proves the two representations of Z_0. Moreover, (24) and $R^\top P_\star R = 0$ give

$$N_0 = \rho_0 R^\top S_\star R \succeq \rho_0 \delta_\star I \succeq \rho_- \delta_\star I \succ 0. \tag{129}$$

It remains to verify the reduced KKT assertion directly. Trace inactivity and original scalar complementarity imply $\nu_\star = 0$. Original dual stationarity in (6) hence gives

$$S_\star = C - \mathcal{A}^* y_\star = -H(y_\star). \tag{130}$$

Because $\bar{P} \in \mathfrak{P}_\star$, one has $\bar{P} X_\star \bar{P} = X_\star$. Thus $X_\star \in \mathcal{D}(\bar{P})$, and primal feasibility yields

$$\begin{aligned} \nabla \Psi_{\rho_0, y_\star}(X_\star) &= \mathcal{A}^*(\mathcal{A}(X_\star) - b) - \rho_0 H(y_\star) \\ &= \rho_0 S_\star. \end{aligned} \tag{131}$$

Consequently,

$$\bar{P} \nabla \Psi_{\rho_0, y_\star}(X_\star) \bar{P} - Z_0 = 0, \tag{132}$$

which is reduced stationarity with trace multiplier zero.

Equation (128) shows $Z_0 \in \mathcal{E}(\bar{P})$ and $Z_0 \succeq 0$. Finally,

$$Z_0 X_\star = \rho_0 \bar{P} S_\star \bar{P} X_\star = \rho_0 \bar{P} S_\star X_\star = 0, \tag{133}$$

and transposition gives $X_\star Z_0 = 0$. Together with $\operatorname{tr} X_\star < 1$, these are all conditions of Proposition E.5 for the triple $(X_\star, 0, Z_0)$. □

Lemma E.7 (Ambient extension of the direct rotation)**.** *For every $\bar{P} \in \mathfrak{P}_r$, there is an open ball $\mathcal{O}_{\bar{P}} \subset \mathbb{S}^n$ centered at $\bar{P}$ on which the following maps are well defined:*

$$\begin{aligned} A_{\bar{P}}(\Pi) &:= \Pi \bar{P} + (I - \Pi)(I - \bar{P}), \\ Q_{\bar{P}}(\Pi) &:= A_{\bar{P}}(\Pi)[A_{\bar{P}}(\Pi)^\top A_{\bar{P}}(\Pi)]^{-1/2}, \\ \widehat{P}_{\bar{P}}(\Pi) &:= Q_{\bar{P}}(\Pi) \bar{P} Q_{\bar{P}}(\Pi)^\top. \end{aligned} \tag{134}$$

They are C^∞, $Q_{\bar{P}}(\Pi) \in O(n)$, and $\widehat{P}_{\bar{P}}(\Pi) \in \mathfrak{P}_r$. If $P \in \mathcal{O}_{\bar{P}} \cap \mathfrak{P}_r$, then

$$Q_{\bar{P}}(P) \bar{P} Q_{\bar{P}}(P)^\top = P. \tag{135}$$

Proof. At $\Pi = \bar{P}$, $A_{\bar{P}}(\bar{P}) = I$, and direct expansion gives

$$A_{\bar{P}}(\Pi) - I = (\Pi - \bar{P})(2\bar{P} - I).$$

Since $2\bar{P} - I$ is orthogonal, $A_{\bar{P}}(\Pi)$ is invertible in a small Frobenius ball about $\bar{P}$. Hence the polar factor in (134) is well defined and orthogonal. The affine map $A_{\bar{P}}$ and the inverse square root on

the positive-definite cone are C^∞; see [22]. Thus both displayed maps are smooth, and $\widehat{P}_{\bar{P}}(\Pi)$ is an orthogonal conjugate of $\bar{P}$.

Shrink $\mathcal{O}_{\bar{P}}$, if necessary, so that $\left\|P-\bar{P}\right\|_2 < 1$ for every actual projector P in the ball. Fix such a P and abbreviate

$$A := P\bar{P} + (I-P)(I-\bar{P}), \qquad Q := A(A^\top A)^{-1/2}. \tag{136}$$

For $x \in \mathbb{R}^n$, write $x = x_1 + x_0$ with $x_1 = \bar{P}x$ and $x_0 = (I-\bar{P})x$. Then

$$Ax = Px_1 + (I-P)x_0, \tag{137}$$

and the two terms on the right lie in the orthogonal spaces Range(P) and Null(P). If $Ax = 0$, both terms vanish. If $x_1 \neq 0$, then $Px_1 = 0$ and $\bar{P}x_1 = x_1$, which would imply

$$\left\|P-\bar{P}\right\|_2 \geq \frac{\left\|(P-\bar{P})x_1\right\|_2}{\|x_1\|_2} = 1. \tag{138}$$

Hence $x_1 = 0$. Similarly, $(I-P)x_0 = 0$ gives $Px_0 = x_0$ and $\bar{P}x_0 = 0$, so the same estimate forces $x_0 = 0$. Therefore A is nonsingular.

Put $B = (A^\top A)^{-1/2}$. Then

$$Q^\top Q = BA^\top AB = I, \tag{139}$$

so $Q \in O(n)$. Direct multiplication also gives

$$PA = P\bar{P} = A\bar{P}, \tag{140}$$

and

$$A^\top A = \bar{P}P\bar{P} + (I-\bar{P})(I-P)(I-\bar{P}). \tag{141}$$

Thus $A^\top A$ commutes with $\bar{P}$, and so does its spectral function B. Equation (140) now yields

$$PQ = PAB = A\bar{P}B = AB\bar{P} = Q\bar{P}. \tag{142}$$

Multiplication by $Q^\top$ gives $P = Q\bar{P}Q^\top$, proving (135). □

Definition E.8 (Strict-complementary graph patch)**.** Define the coordinate domain

$$\mathcal{C}_{\bar{P}} := \mathbb{R}^{(r-s)\times s} \times \mathbb{S}^s_{++} \times \mathbb{S}^{r-s}_{++}, \tag{143}$$

and the pair set

$$\begin{aligned} \mathcal{M}^{\mathrm{graph}}_{\bar{P}} := \{(X,Z) \in \mathcal{E}(\bar{P}) \times \mathcal{E}(\bar{P}) : X \succeq 0,\ Z \succeq 0,\ XZ = 0,\\ \operatorname{rank} X = s,\ \operatorname{rank} Z = r-s,\\ U_\star^\top X U_\star \succ 0,\ R^\top Z R \succ 0\}. \end{aligned} \tag{144}$$

The last two strict inequalities select the graph chart based at $[U_\star, R]$.

Lemma E.9 (Graph coordinates are a smooth bijection)**.** *For $(T,\Sigma,N) \in \mathcal{C}_{\bar{P}}$, define*

$$\begin{aligned} X(T,\Sigma) &:= (U_\star + RT)\Sigma(U_\star + RT)^\top,\\ Z(T,N) &:= (R - U_\star T^\top)N(R - U_\star T^\top)^\top. \end{aligned} \tag{145}$$

and

$$\Gamma_{\bar{P}}(T,\Sigma,N) := (X(T,\Sigma), Z(T,N)).$$

Then $\Gamma_{\bar{P}} : \mathcal{C}_{\bar{P}} \to \mathcal{M}^{\mathrm{graph}}_{\bar{P}}$ is a C^∞ bijection with C^∞ inverse

$$\Gamma^{-1}_{\bar{P}}(X,Z) = \left(R^\top X U_\star (U_\star^\top X U_\star)^{-1}, U_\star^\top X U_\star, R^\top Z R\right). \tag{146}$$

Proof. Step 1: the forward map enters the graph patch. Write $G(T) = U_\star + RT$ and $J(T) = R - U_\star T^\top$. Since $\bar{P}U_\star = U_\star$ and $\bar{P}R = R$, both column spaces lie in $\mathrm{Range}(\bar{P})$, so $X(T,\Sigma), Z(T,N) \in \mathcal{E}(\bar{P})$. Moreover,

$$G(T)^\top G(T) = I + T^\top T \succ 0, \qquad J(T)^\top J(T) = I + TT^\top \succ 0, \qquad G(T)^\top J(T) = 0.$$

Thus $G(T)$ and $J(T)$ have full column rank. Since $\Sigma \succ 0$ and $N \succ 0$, the two matrices are PSD and have ranks s and $r - s$. The identity $G(T)^\top J(T) = 0$ gives $X(T,\Sigma)Z(T,N) = 0$. Their blocks in the basis $[U_\star, R]$ are

$$\begin{bmatrix} \Sigma & \Sigma T^\top \\ T\Sigma & T\Sigma T^\top \end{bmatrix}, \qquad \begin{bmatrix} T^\top N T & -T^\top N \\ -NT & N \end{bmatrix}. \tag{147}$$

The upper-left block of the first matrix is $\Sigma \succ 0$, and the lower-right block of the second is $N \succ 0$. Hence the forward map is well defined into $\mathcal{M}_{\bar{P}}^{\mathrm{graph}}$.

Step 2: recover Σ, T, and the entire matrix X. Let $(X, Z) \in \mathcal{M}_{\bar{P}}^{\mathrm{graph}}$ and write

$$\bar{V}^\top X \bar{V} = \begin{bmatrix} X_{11} & X_{12} \\ X_{12}^\top & X_{22} \end{bmatrix}. \tag{148}$$

The graph-patch condition gives $X_{11} \succ 0$. Define

$$\Sigma := X_{11}, \qquad T := X_{12}^\top X_{11}^{-1}. \tag{149}$$

The block elimination matrix

$$L := \begin{bmatrix} I & 0 \\ -X_{12}^\top X_{11}^{-1} & I \end{bmatrix}$$

is nonsingular and satisfies

$$L(\bar{V}^\top X \bar{V})L^\top = \begin{bmatrix} X_{11} & 0 \\ 0 & X_{22} - X_{12}^\top X_{11}^{-1} X_{12} \end{bmatrix}. \tag{150}$$

Consequently,

$$\mathrm{rank}\, X = s + \mathrm{rank}(X_{22} - X_{12}^\top X_{11}^{-1} X_{12}).$$

Since $\mathrm{rank}\, X = s$, the Schur complement in (150) vanishes. Equations (149) therefore give

$$X_{12} = \Sigma T^\top, \qquad X_{22} = T\Sigma T^\top,$$

so comparison with (147) proves $X = X(T,\Sigma)$. The recovery is unique: the $(1,1)$ block fixes Σ, and the $(2,1)$ block fixes $T = (R^\top X U_\star)\Sigma^{-1}$.

Step 3: recover N and the entire matrix Z. Write

$$\bar{V}^\top Z \bar{V} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{12}^\top & Z_{22} \end{bmatrix}, \qquad N := Z_{22} = R^\top Z R \succ 0. \tag{151}$$

We already know $X = G(T)\Sigma G(T)^\top$, where both $G(T)$ and Σ have full rank. From $XZ = 0$,

$$G(T)\Sigma G(T)^\top Z = 0.$$

Left multiplication first by $(G^\top G)^{-1}G^\top$ and then by Σ^{-1} gives $G(T)^\top Z = 0$. By symmetry, $ZG(T) = 0$ as well. In the $\bar{V}$ coordinates, the latter identity is

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{12}^\top & N \end{bmatrix} \begin{bmatrix} I \\ T \end{bmatrix} = 0. \tag{152}$$

Its lower block yields $Z_{12}^\top = -NT$, hence $Z_{12} = -T^\top N$; the upper block then yields $Z_{11} = T^\top NT$. Thus $Z = Z(T, N)$, and N is unique because it is the fixed block $R^\top ZR$.

Step 4: mutual inverses and smoothness. For a forward image, (147) gives

$$U_\star^\top XU_\star = \Sigma, \qquad R^\top XU_\star = T\Sigma, \qquad R^\top ZR = N,$$

so (146) recovers (T, Σ, N). Conversely, Steps 2–3 prove that the coordinates in (146) reconstruct the original pair. Hence the two maps are mutual inverses. The forward map is polynomial. The inverse consists of linear compressions and inversion of the positive-definite matrix $U_\star^\top XU_\star$; it is therefore C^∞ on the stated domain. □

Corollary E.10 (The base pair lies in the interior of the graph patch)**.** *The pair $(X_\star, Z_0)$ equals $(X(0, \Sigma_\star), Z(0, N_0))$ and belongs to $\mathcal{M}_{\bar{P}}^{\mathrm{graph}}$. Every sufficiently close PSD complementary pair in $\mathcal{E}(\bar{P})$ having ranks $(s, r-s)$ also belongs to the same graph patch.*

Proof. The first assertion follows from Lemma E.6. At the base pair, $U_\star^\top X_\star U_\star = \Sigma_\star \succ 0$ and $R^\top Z_0 R = N_0 \succ 0$. These strict inequalities persist under small perturbations by Weyl's inequality. □

Definition E.11 (Pullback to a fixed reduced space)**.** For $\theta = (\rho, y, \Pi)$, put $Q = Q_{\bar{P}}(\Pi)$ and define on $\mathcal{E}(\bar{P})$

$$\widetilde{\Psi}_{\rho,y,\Pi}(\widetilde{X}) := \Psi_{\rho,y}(Q\widetilde{X}Q^\top). \tag{153}$$

If $\Pi = P \in \mathfrak{P}_r$, orthogonal conjugation by $Q_{\bar{P}}(P)$ maps $\mathcal{D}(\bar{P})$ bijectively onto $\mathcal{D}(P)$, because it preserves PSD and trace and (135) identifies the two support projectors.

Lemma E.12 (Gradient of the pulled-back objective)**.** *The gradient of (153) in the fixed Euclidean space $\mathcal{E}(\bar{P})$ is*

$$G(\widetilde{X}; \theta) := \bar{P}Q^\top \big[\mathcal{A}^*(\mathcal{A}(Q\widetilde{X}Q^\top) - b) - \rho H(y)\big] Q\bar{P}, \qquad Q = Q_{\bar{P}}(\Pi). \tag{154}$$

Proof. For $\Delta\widetilde{X} \in \mathcal{E}(\bar{P})$, the chain rule and cyclicity of the trace give

$$D\widetilde{\Psi}[\Delta\widetilde{X}] = \left\langle Q^\top \nabla\Psi_{\rho,y}(Q\widetilde{X}Q^\top)Q, \Delta\widetilde{X} \right\rangle.$$

Since $\Delta\widetilde{X} = \bar{P}\Delta\widetilde{X}\bar{P}$, inserting $\bar{P}$ on both sides of the first argument does not change the inner product. Substitution of (125) gives (154). Smoothness in every argument follows from Lemma E.7 and multilinearity of the remaining operations. □

Definition E.13 (Fixed-space graph KKT map)**.** The reduced stationarity equation in graph coordinates is

$$\mathscr{F}(\Sigma, T, N; \rho, y, \Pi) := \bar{V}^\top [G(X(T, \Sigma); \theta) - Z(T, N)]\bar{V} = 0. \tag{155}$$

Its unknown space $\mathbb{S}^s \times \mathbb{R}^{(r-s)\times s} \times \mathbb{S}^{r-s}$ and value space $\mathbb{S}^r$ have the same dimension $r(r+1)/2$. Denote the base variables by

$$w_0 = (\Sigma_\star, 0, N_0), \qquad \theta_0 = (\rho_0, y_\star, \bar{P}).$$

Lemma E.14 (The graph KKT map has the required base zero)**.** *The point (w_0, θ_0) satisfies*

$$\mathscr{F}(w_0; \theta_0) = 0. \tag{156}$$

Proof. At $\Pi = \bar{P}$, the direct rotation is the identity. The graph formulas give $X(0, \Sigma_\star) = X_\star$ and $Z(0, N_0) = Z_0$. By Lemma E.6, $\bar{P}\nabla\Psi_{\rho_0, y_\star}(X_\star)\bar{P} = Z_0$. Thus $G(X_\star; \theta_0) - Z_0 = 0$ in $\mathcal{E}(\bar{P})$, and compression by $\bar{V}$ proves (156). □

Theorem E.15 (Nonsingularity of the reduced KKT Jacobian)**.** *The partial derivative $D_w\mathscr{F}(w_0; \theta_0)$ is a linear isomorphism.*

Proof. Take a direction $(\Delta\Sigma, \Delta T, \Delta N)$ in the kernel. Direct differentiation of the graph formulas at $(\Sigma_\star, 0, N_0)$ gives

$$\Delta X = U_\star \Delta\Sigma U_\star^\top + R\Delta T\Sigma_\star U_\star^\top + U_\star\Sigma_\star\Delta T^\top R^\top, \tag{157}$$

$$\Delta Z = R\Delta N R^\top - U_\star\Delta T^\top N_0 R^\top - RN_0\Delta T U_\star^\top. \tag{158}$$

At θ_0, the direct rotation is the identity, and differentiation of the pulled-back gradient in its matrix argument gives $\bar{P}\mathcal{A}^*\mathcal{A}(\Delta X)\bar{P}$. Since $Y \mapsto \bar{V}^\top Y \bar{V}$ is an isomorphism from $\mathcal{E}(\bar{P})$ onto $\mathbb{S}^r$, the kernel condition is equivalent to

$$\bar{P}\mathcal{A}^*\mathcal{A}(\Delta X)\bar{P} - \Delta Z = 0. \tag{159}$$

In the orthonormal basis $\bar{V} = [U_\star, R]$, equations (157)–(158) read

$$\bar{V}^\top \Delta X \bar{V} = \begin{bmatrix} \Delta\Sigma & \Sigma_\star\Delta T^\top \\ \Delta T\Sigma_\star & 0 \end{bmatrix}, \qquad \bar{V}^\top \Delta Z \bar{V} = \begin{bmatrix} 0 & -\Delta T^\top N_0 \\ -N_0\Delta T & \Delta N \end{bmatrix}. \tag{160}$$

Taking the Frobenius inner product of (159) with ΔX gives

$$0 = \|\mathcal{A}(\Delta X)\|_2^2 - \langle \Delta Z, \Delta X\rangle. \tag{161}$$

The diagonal blocks in (160) contribute zero to $\langle \Delta Z, \Delta X\rangle$. The two off-diagonal products give

$$\begin{aligned} \langle \Delta Z, \Delta X\rangle &= -2\operatorname{tr}(N_0\Delta T\Sigma_\star\Delta T^\top) \\ &= -2\left\|N_0^{1/2}\Delta T\Sigma_\star^{1/2}\right\|_F^2. \end{aligned} \tag{162}$$

Substitution into (161) yields the energy identity

$$0 = \|\mathcal{A}(\Delta X)\|_2^2 + 2\left\|N_0^{1/2}\Delta T\Sigma_\star^{1/2}\right\|_F^2. \tag{163}$$

Both terms are nonnegative. Because $N_0 \succ 0$ and $\Sigma_\star \succ 0$, the second term vanishes only when $\Delta T = 0$; the first simultaneously gives $\mathcal{A}(\Delta X) = 0$. Equation (157) now reduces to

$$\Delta X = U_\star\Delta\Sigma U_\star^\top, \qquad \mathcal{A}(U_\star\Delta\Sigma U_\star^\top) = 0. \tag{164}$$

The active-block injectivity in Lemma E.2 therefore forces $\Delta\Sigma = 0$, and hence $\Delta X = 0$. Equation (159) then gives $\Delta Z = 0$. With $\Delta T = 0$, (158) becomes $\Delta Z = R\Delta N R^\top$; multiplication by $R^\top$ and R yields $\Delta N = 0$. Thus the kernel is trivial.

Finally,

$$\begin{aligned} \dim\mathbb{S}^s + s(r-s) + \dim\mathbb{S}^{r-s} &= \frac{s(s+1)}{2} + s(r-s) + \frac{(r-s)(r-s+1)}{2} \\ &= \frac{r(r+1)}{2} = \dim\mathbb{S}^r. \end{aligned}$$

Hence the injective Jacobian is a linear isomorphism. □

Lemma E.16 (Local uniqueness of a convex minimizer is global uniqueness)**.** *Let f be convex on a convex set $\mathcal{C}$. If $x^\sharp$ minimizes f over $\mathcal{C}$ and is the only minimizer in some relative neighborhood of itself, then it is the unique global minimizer.*

Proof. If a distinct global minimizer $\widetilde{x}$ existed, convexity would give, for every $t \in [0,1]$,

$$f((1-t)x^\sharp + t\widetilde{x}) \le (1-t)f(x^\sharp) + tf(\widetilde{x}) = f(x^\sharp). \tag{165}$$

Minimality forces equality. For arbitrarily small $t > 0$ these distinct minimizers lie in every relative neighborhood of $x^\sharp$, a contradiction. □

Theorem E.17 (A C^∞ reduced KKT branch at a fixed base pair)**.** *For the fixed base pair $\rho_0 \in [\rho_-, \rho_+]$ and $\bar{P} \in \mathfrak{P}_\star$, there exist*

- *an open interval $I_{\rho_0} \subset (0,\infty)$ containing ρ_0;*
- *an open neighborhood $\mathcal{Y}_{\rho_0,\bar{P}} \subset \mathbb{R}^m$ containing $y_\star$;*
- *an open neighborhood $\mathcal{O}_{\rho_0,\bar{P}} \subset \mathcal{O}_{\bar{P}}$ containing $\bar{P}$;*
- *two C^∞ ambient maps*

$$\mathcal{X}_{\rho_0,\bar{P}}, \mathcal{Z}_{\rho_0,\bar{P}} : I_{\rho_0} \times \mathcal{Y}_{\rho_0,\bar{P}} \times \mathcal{O}_{\rho_0,\bar{P}} \longrightarrow \mathbb{S}^n.$$

Writing $\theta = (\rho, y, \Pi)$ and $Q = Q_{\bar{P}}(\Pi)$, these maps have the form

$$\mathcal{X}_{\rho_0,\bar{P}}(\theta) := QX(T(\theta), \Sigma(\theta))Q^\top, \qquad \mathcal{Z}_{\rho_0,\bar{P}}(\theta) := QZ(T(\theta), N(\theta))Q^\top. \tag{166}$$

For every

$$(\rho, y, P) \in I_{\rho_0} \times \mathcal{Y}_{\rho_0,\bar{P}} \times (\mathcal{O}_{\rho_0,\bar{P}} \cap \mathfrak{P}_r),$$

set $X^\sharp = \mathcal{X}_{\rho_0,\bar{P}}(\rho, y, P)$ and $Z^\sharp = \mathcal{Z}_{\rho_0,\bar{P}}(\rho, y, P)$. Then:

(i) $X^\sharp$ is the unique global minimizer of $\min_{X \in \mathcal{D}(P)} \Psi_{\rho,y}(X)$, and hence $X^\sharp = X^\sharp_\rho(y, P)$;

(ii) $(X^\sharp, Z^\sharp)$ satisfies the trace-inactive reduced KKT system

$$X^\sharp, Z^\sharp \in \mathcal{E}(P), \quad X^\sharp, Z^\sharp \succeq 0, \quad X^\sharp Z^\sharp = 0,$$

$$P\nabla\Psi_{\rho,y}(X^\sharp)P = Z^\sharp, \qquad \operatorname{tr} X^\sharp < 1,$$

and $Z^\sharp$ is the unique trace-inactive reduced slack;

(iii) $\operatorname{rank} X^\sharp = s$ and $\operatorname{rank} Z^\sharp = r - s$;

(iv) at the base point,

$$\mathcal{X}_{\rho_0,\bar{P}}(\rho_0, y_\star, \bar{P}) = X_\star, \qquad \mathcal{Z}_{\rho_0,\bar{P}}(\rho_0, y_\star, \bar{P}) = Z_0.$$

For a nonprojector third argument Π, the two maps are only smooth ambient extensions and need not represent an optimization problem.

Proof. For readability, suppress the fixed base-pair subscripts on the local branch maps throughout the proof. *Step 1: apply the Euclidean implicit-function theorem.* The map $\mathscr{F}$ is C^∞ on an open product of finite-dimensional Euclidean spaces. Its base value is zero by Lemma E.14, and its

unknown Jacobian is invertible by Theorem E.15. The classical implicit-function theorem [3] supplies open neighborhoods and a unique local C^∞ map

$$w(\theta) = (\Sigma(\theta), T(\theta), N(\theta)), \qquad \mathscr{F}(w(\theta); \theta) = 0. \tag{167}$$

The graph-coordinate unknown space allows arbitrary symmetric Σ, N, but positive definiteness is open. Since $\Sigma(\theta_0) = \Sigma_\star \succ 0$ and $N(\theta_0) = N_0 \succ 0$, the parameter neighborhood can be shrunk so that

$$\Sigma(\theta) \succ 0, \qquad N(\theta) \succ 0 \tag{168}$$

throughout it. Continuity of the trace and $\operatorname{tr} X_\star < 1$ allow the simultaneous condition

$$\operatorname{tr} X(T(\theta), \Sigma(\theta)) < 1. \tag{169}$$

Step 2: recover complementary PSD matrices of stable rank. By (145) and (168), the pulled-back pair

$$\widetilde{X}(\theta) = X(T(\theta), \Sigma(\theta)), \qquad \widetilde{Z}(\theta) = Z(T(\theta), N(\theta)) \tag{170}$$

is PSD, complementary, and has ranks s and $r - s$. Orthogonal conjugation by $Q = Q_{\bar{P}}(\Pi)$ preserves these properties, proving the same assertions for the ambient maps in (166). Equation (169) sets the trace multiplier equal to zero.

Step 3: identify the branch with the actual reduced optimizer. Suppose now that the third parameter P is an actual rank-r projector. By Lemma E.7, $Q\bar{P}Q^\top = P$, and hence orthogonal conjugation maps $\mathcal{D}(\bar{P})$ bijectively onto $\mathcal{D}(P)$. Equation (167) is the pulled-back stationarity identity

$$G(\widetilde{X}; \rho, y, P) = \widetilde{Z}. \tag{171}$$

Multiplication on the left by Q and on the right by $Q^\top$ yields

$$P \nabla \Psi_{\rho,y}(\mathcal{X}(\theta)) P = \mathcal{Z}(\theta). \tag{172}$$

Together with PSD feasibility, complementarity, and trace inactivity, this is the complete reduced KKT system of Proposition E.5. Convexity therefore proves that $\mathcal{X}(\theta)$ is a global optimizer.

Step 4: prove local uniqueness among all reduced KKT pairs. Choose once, at the base point, a matrix neighborhood in which the positive eigenvalues of $X_\star$ and Z_0 stay bounded away from zero under perturbation. Shrink the parameter neighborhood so that the branch pair remains inside half of this matrix neighborhood. Let $\widetilde{X}$ be any other optimizer inside the full neighborhood. Its trace is inactive by closeness to $X_\star$, and its KKT slack is forced by stationarity to be

$$\widetilde{Z} = P \nabla \Psi_{\rho,y}(\widetilde{X}) P. \tag{173}$$

The right-hand side depends continuously on $(\rho, y, P, \widetilde{X})$, so $\widetilde{Z}$ is close to Z_0 whenever the parameters and $\widetilde{X}$ are close to their base values. Weyl's inequality then gives

$$\operatorname{rank} \widetilde{X} \geq s, \qquad \operatorname{rank} \widetilde{Z} \geq r - s. \tag{174}$$

PSD complementarity inside the r-dimensional space $\operatorname{Range}(P)$ gives

$$\operatorname{rank} \widetilde{X} + \operatorname{rank} \widetilde{Z} \leq r. \tag{175}$$

Thus equality holds throughout (174)–(175); the ranks are exactly $(s, r - s)$. The strict active blocks that define the graph patch are open conditions and hold at $(X_\star, Z_0)$, so the same matrix

neighborhood places the pulled-back competitor pair in $\mathcal{M}_{\bar{P}}^{\mathrm{graph}}$. By Lemma E.9, it has unique coordinates $(\widetilde{\Sigma}, \widetilde{T}, \widetilde{N})$, and its stationarity is the same equation

$$\mathscr{F}(\widetilde{\Sigma}, \widetilde{T}, \widetilde{N}; \theta) = 0. \tag{176}$$

Let $\mathcal{W}$ be the unknown-variable neighborhood supplied by the implicit-function theorem. By continuity of $Q_{\bar{P}}$ and of the graph inverse (146), shrink the matrix and parameter neighborhoods once more so that every competitor in the smaller matrix neighborhood has pulled-back graph coordinates $(\widetilde{\Sigma}, \widetilde{T}, \widetilde{N}) \in \mathcal{W}$. After a further parameter shrink, the branch pair remains in half of the resulting matrix neighborhood. Local uniqueness in the implicit-function theorem now forces these coordinates to equal $w(\theta)$. Hence the branch optimizer is locally unique.

Step 5: upgrade local uniqueness to global uniqueness. The feasible set $\mathcal{D}(P)$ is convex and $\Psi_{\rho,y}$ is convex. Lemma E.16 therefore excludes any second optimizer outside the matrix neighborhood. The base identities follow from $w(\theta_0) = w_0$ and $Q_{\bar{P}}(\bar{P}) = I$. For a nonprojector Π, only the smooth ambient construction is asserted, because no actual moving reduced feasible set is attached to Π. □

Lemma E.18 (Smooth local branches have quantitative $C^{1,1}$ margins)**.** *For every base pair $(\rho_0, \bar{P})$, the parameter neighborhood in Theorem E.17 can be shrunk to an open Euclidean ball $\mathcal{U}'_{\rho_0,\bar{P}}$ whose closure is compact and contained in the original branch domain. On that closure there are finite constants $M_{X,1}, M_{X,2}, M_{Z,1}, M_{Z,2}$ such that*

$$\begin{aligned} &\sup_{\theta\in\overline{\mathcal{U}'}} \|D\mathcal{X}(\theta)\|_{\mathrm{op}} \le M_{X,1}, \quad \sup_{\theta\in\overline{\mathcal{U}'}} \left\|D^2\mathcal{X}(\theta)\right\|_{\mathrm{op}} \le M_{X,2}, \\ &\sup_{\theta\in\overline{\mathcal{U}'}} \|D\mathcal{Z}(\theta)\|_{\mathrm{op}} \le M_{Z,1}, \quad \sup_{\theta\in\overline{\mathcal{U}'}} \left\|D^2\mathcal{Z}(\theta)\right\|_{\mathrm{op}} \le M_{Z,2}. \end{aligned} \tag{177}$$

In particular,

$$\|D\mathcal{X}(\theta_1) - D\mathcal{X}(\theta_2)\|_{\mathrm{op}} \le M_{X,2} \|\theta_1 - \theta_2\|, \tag{178}$$

and analogously for $D\mathcal{Z}$, for all $\theta_1, \theta_2 \in \mathcal{U}'$.

Proof. Choose a sufficiently small open ball centered at the base parameter so that its closure remains in the original open branch domain. In finite dimensions this closure is compact. Since $\mathcal{X}, \mathcal{Z} \in C^\infty$, the four functions obtained by taking the operator norms of their first and second derivatives are continuous and attain finite maxima on that closure; these maxima are the constants in (177).

For $\theta_1, \theta_2 \in \mathcal{U}'$, convexity of the ball keeps $\theta(t) = \theta_2 + t(\theta_1 - \theta_2)$ in $\mathcal{U}'$. Apply the fundamental theorem of calculus to the operator-valued map $D\mathcal{X}$:

$$D\mathcal{X}(\theta_1) - D\mathcal{X}(\theta_2) = \int_0^1 D^2\mathcal{X}(\theta(t))[\theta_1 - \theta_2]\, dt. \tag{179}$$

Taking operator norms and using (177) proves (178). The proof for $D\mathcal{Z}$ is identical. □

E.2 Uniformization over the redundant projector family

Lemma E.19 (Invariance of the base-projector manifold)**.** *If $P \in \mathfrak{P}_\star$, then*

$$P^+(y_\star, P, \eta) \in \mathfrak{P}_\star \qquad \text{for every } \eta \in \mathbb{R}. \tag{180}$$

Proof. Choose $V \in \mathrm{St}(n,r)$ with $VV^\top = P$ and put $C = V^\top U_\star$, so $VC = U_\star$. By Lemma 5.1, $H(y_\star)U_\star = 0$. For $Y = V + \eta(I-P)H(y_\star)V$,

$$YC = U_\star + \eta(I-P)H(y_\star)U_\star = U_\star.$$

Thus $\mathrm{Range}(U_\star) \subseteq \mathrm{Range}(Y) = \mathrm{Range}(P^+)$, and Lemma 4.4 gives $P^+ \in \mathfrak{P}_r$. □

Definition E.20 (Ambient projector update and mismatch map)**.** For a base point $a = (\rho_0, y_\star, \bar{P})$, choose $\bar{V}_a \bar{V}_a^\top = \bar{P}$, write

$$Q_a(\Pi) := Q_{\bar{P}}(\Pi), \quad \widehat{P}_a(\Pi) := Q_a(\Pi)\bar{P}Q_a(\Pi)^\top, \quad V_a(\Pi) := Q_a(\Pi)\bar{V}_a,$$

and define

$$D_a(y,\Pi) := (I - \widehat{P}_a(\Pi))H(y)V_a(\Pi), \qquad Y_a(y,\Pi,\eta) := V_a(\Pi) + \eta D_a(y,\Pi).$$

The ambient full-residual update is

$$\mathcal{P}_a(y,\Pi,\eta) := Y_a(Y_a^\top Y_a)^{-1}Y_a^\top. \tag{181}$$

It is C^∞, since $Y_a^\top Y_a = I + \eta^2 D_a^\top D_a \succ 0$. For an actual projector P, $\mathcal{P}_a(y,P,\eta) = P^+(y,P,\eta)$.

Let $\mathcal{U}_a$ be the domain of the local branch $\mathcal{X}_a$. The natural mismatch domain is the open set

$$\mathcal{D}_a^G := \{(\rho,y,\Pi,\eta) : (\rho,y,\Pi) \in \mathcal{U}_a,\ (\rho,y,\mathcal{P}_a(y,\Pi,\eta)) \in \mathcal{U}_a\}.$$

On this domain define

$$\begin{aligned}\mathcal{G}_a(\rho,y,\Pi,\eta) := \mathcal{P}_a(y,\Pi,\eta)\mathcal{X}_a(\rho,y,\Pi)\mathcal{P}_a(y,\Pi,\eta) \\ - \mathcal{X}_a(\rho,y,\mathcal{P}_a(y,\Pi,\eta)).\end{aligned} \tag{182}$$

Both target evaluations are therefore part of the definition of the domain, and $\mathcal{G}_a$ is C^∞.

The base set

$$\mathcal{K}_0 := [\rho_-,\rho_+] \times \{y_\star\} \times \mathfrak{P}_\star \tag{183}$$

is compact.

Lemma E.21 (Fix final margins before extracting a finite chart cover)**.** *For every $a \in \mathcal{K}_0$, one can first choose final numbers $r_a, \eta_a > 0$ such that*

$$\overline{B}_{4r_a}(a) \subset \mathcal{U}_a, \qquad \overline{B}_{4r_a}(a) \times [-\eta_a,\eta_a] \subset \mathcal{D}_a^G.$$

There are then finitely many base points $a_1,\dots,a_J$ whose already fixed balls $B_{r_j}(a_j)$ cover $\mathcal{K}_0$. Define

$$\delta_{\mathrm{chart}} := \min_j r_j > 0, \qquad \eta_{\mathrm{chart}} := \min_j \eta_j > 0. \tag{184}$$

Proof. Fix $a \in \mathcal{K}_0$. The local branch domain $\mathcal{U}_a$ is open and contains a. Hence there is $\epsilon_a^X > 0$ such that

$$\overline{B}_{\epsilon_a^X}(a) \subset \mathcal{U}_a. \tag{185}$$

The natural mismatch domain $\mathcal{D}_a^G$ is open and contains $(a,0)$. Therefore there are $\epsilon_a^G, \widehat{\eta}_a > 0$ such that

$$\overline{B}_{\epsilon_a^G}(a) \times [-\widehat{\eta}_a,\widehat{\eta}_a] \subset \mathcal{D}_a^G. \tag{186}$$

Choose, before extracting any cover,

$$0 < r_a < \frac{1}{4}\min\{\epsilon_a^X, \epsilon_a^G\}, \qquad 0 < \eta_a < \widehat{\eta}_a. \tag{187}$$

Equations (185)– (187) give exactly the two closed-set inclusions in the statement. This order is important: the later finite subcover selects among final chart margins rather than among provisional neighborhoods that might subsequently be shrunk.

The family $\{B_{r_a}(a) : a \in \mathcal{K}_0\}$ is an open cover of the compact set $\mathcal{K}_0$, so choose a finite subcover with centers $a_1, \dots, a_J$. Since every selected r_j and η_j is positive, their finite minima in (184) are positive.

□

Definition E.22 (Uniform selected-chart constants)**.** On the compact work sets $K_j := \overline{B}_{4r_j}(a_j)$ and $K_j^G := K_j \times [-\eta_j, \eta_j]$, define

$$\begin{aligned} L_{X,1} &:= \max_j \sup_{\theta\in K_j} \|D\mathcal{X}_j(\theta)\|_{\mathrm{op}}, \quad L_{X,11} := \max_j \sup_{\theta\in K_j} \left\|D^2\mathcal{X}_j(\theta)\right\|_{\mathrm{op}}, \\ L_{Z,1} &:= \max_j \sup_{\theta\in K_j} \|D\mathcal{Z}_j(\theta)\|_{\mathrm{op}}, \quad L_{Z,11} := \max_j \sup_{\theta\in K_j} \left\|D^2\mathcal{Z}_j(\theta)\right\|_{\mathrm{op}}. \end{aligned} \tag{188}$$

and

$$M_{P^+} := \max_j \sup_{K_j^G} \|\partial_\eta \mathcal{P}_j\|_F, \qquad L_G := \max_j \sup_{K_j^G} \left\|D^2\mathcal{G}_j\right\|_{\mathrm{op}}. \tag{189}$$

Every quantity is finite by smoothness and compactness. The constants $L_{X,11}$ and L_G are also Lipschitz constants for $D\mathcal{X}_j$ and $D\mathcal{G}_j$ on the corresponding convex work sets.

Corollary E.23 (Uniform target and slack spectral bounds)**.** *There exist $r_{\mathrm{trg}} > 0$ and $0 < \underline{\sigma} \le \overline{\sigma} < \infty$, $\underline{\nu} > 0$ such that every actual parameter satisfying*

$$\rho \in [\rho_-, \rho_+], \qquad \|y - y_\star\|_2 < r_{\mathrm{trg}}, \qquad P \in \mathfrak{P}_r, \qquad \mathrm{dist}_F(P, \mathfrak{P}_\star) < r_{\mathrm{trg}}$$

belongs, together with a base point $(\rho, y_\star, \bar{P})$, to one selected chart. The unique target and slack admit factorizations

$$X_\rho^\sharp(y, P) = U\Sigma U^\top, \quad Z_\rho^\sharp(y, P) = R_c N R_c^\top, \quad \underline{\sigma} I \preceq \Sigma \preceq \overline{\sigma} I, \quad N \succeq \underline{\nu} I \tag{190}$$

Proof. Fix $r_{\mathrm{trg}} > 0$ with $2r_{\mathrm{trg}} < \delta_{\mathrm{chart}}$, and define the closed target tube

$$\mathcal{T}_{\mathrm{trg}} := \left\{ (\rho, y, P) : \begin{array}{cc} \rho \in [\rho_-, \rho_+], & P \in \mathfrak{P}_r, \\ \|y - y_\star\|_2 \le r_{\mathrm{trg}}, & \mathrm{dist}_F(P, \mathfrak{P}_\star) \le r_{\mathrm{trg}} \end{array} \right\}. \tag{191}$$

This set is compact because $[\rho_-, \rho_+]$ and $\mathfrak{P}_r$ are compact, y lies in a closed bounded ball, and both distance constraints are closed. For any $(\rho, y, P) \in \mathcal{T}_{\mathrm{trg}}$, choose $\bar{P} \in \mathfrak{P}_\star$ attaining $\mathrm{dist}_F(P, \mathfrak{P}_\star)$; existence follows from Lemma E.3. Set

$$a = (\rho, y_\star, \bar{P}), \qquad \theta = (\rho, y, P). \tag{192}$$

Then $a \in \mathcal{K}_0$ and

$$\|\theta - a\| \le \|y - y_\star\|_2 + \left\|P - \bar{P}\right\|_F \le 2r_{\mathrm{trg}} < \delta_{\mathrm{chart}}. \tag{193}$$

Select j from the finite cover in Lemma E.21 so that $a \in B_{r_j}(a_j)$. Then

$$\begin{aligned}\|\theta - a_j\| &\le \|\theta - a\| + \|a - a_j\| \\ &< \delta_{\text{chart}} + r_j \le 2r_j. \end{aligned} \tag{194}$$

Both endpoints therefore lie in the convex ball $B_{2r_j}(a_j)$. Here

$$[a, \theta] := \{(1-t)a + t\theta : 0 \le t \le 1\}$$

denotes the straight line segment joining a and θ in the ambient parameter space. Convexity gives

$$[a, \theta] \subset B_{2r_j}(a_j) \subset K_j = \overline{B}_{4r_j}(a_j) \subset \mathcal{U}_j. \tag{195}$$

The actual-projector identification in Theorem E.17 then gives the unique target and slack.

For each selected chart $j \in \{1, \dots, J\}$, the set $K_j \cap \mathcal{T}_{\text{trg}}$ is compact and nonempty: it contains the chart center $a_j \in \mathcal{K}_0$. Every parameter in this intersection has an actual rank-r projector as its third component and lies in $K_j \subset \mathcal{U}_j$. By Theorem E.17, the continuous branch matrices $\mathcal{X}_j(\theta)$ and $\mathcal{Z}_j(\theta)$ are PSD with fixed ranks s and $r - s$, respectively. Their positive eigenvalues are continuous on this compact set. Thus define explicitly

$$\begin{aligned}\underline{\sigma}_j &:= \min_{\theta \in K_j \cap \mathcal{T}_{\text{trg}}} \lambda_s(\mathcal{X}_j(\theta)) > 0, \\ \overline{\sigma}_j &:= \max_{\theta \in K_j \cap \mathcal{T}_{\text{trg}}} \lambda_1(\mathcal{X}_j(\theta)) < \infty, \\ \underline{\nu}_j &:= \min_{\theta \in K_j \cap \mathcal{T}_{\text{trg}}} \lambda_{r-s}(\mathcal{Z}_j(\theta)) > 0. \end{aligned}$$

Finally define

$$\underline{\sigma} := \min_j \underline{\sigma}_j, \qquad \overline{\sigma} := \max_j \overline{\sigma}_j, \qquad \underline{\nu} := \min_j \underline{\nu}_j. \tag{196}$$

Only finitely many charts occur, so the first and third constants are strictly positive and the second is finite. These choices prove (190) uniformly on the closed tube $\mathcal{T}_{\text{trg}}$ in (191). The same bounds remain valid after any later decrease of r_{trg}, since this only restricts the tube. □

Lemma E.24 (Closeness of rank-s PSD matrices implies support closeness)**.** *Let $U_\star, U \in \mathrm{St}(n, s)$, $\Sigma_\star, \Sigma \in \mathbb{S}^s_{++}$, $X_\star = U_\star \Sigma_\star U_\star^\top$, and $X = U\Sigma U^\top$. Suppose $\Sigma \succeq \underline{\sigma} I$. Then*

$$\left\| UU^\top - P_\star \right\|_F \le \frac{\sqrt{2}}{\underline{\sigma}} \left\| X - X_\star \right\|_F. \tag{197}$$

Proof. Since $(I - P_\star)X_\star = 0$,

$$(I - P_\star)X = (I - P_\star)(X - X_\star), \tag{198}$$

and multiplication by an orthogonal projector is nonexpansive. Hence

$$\left\| (I - P_\star)X \right\|_F \le \left\| X - X_\star \right\|_F. \tag{199}$$

On the other hand,

$$\begin{aligned}\|(I - P_\star)X\|_F^2 &= \left\| (I - P_\star)U\Sigma U^\top \right\|_F^2 \\ &= \|(I - P_\star)U\Sigma\|_F^2 \end{aligned}$$

$$= \operatorname{tr}\Big(\Sigma U^\top (I - P_\star) U \Sigma\Big)$$
$$= \operatorname{tr}\Big(U^\top (I - P_\star) U \Sigma^2\Big). \tag{200}$$

Both factors in the last trace are PSD and $\Sigma^2 \succeq \underline{\sigma}^2 I$, so

$$\|(I - P_\star)X\|_F^2 \geq \underline{\sigma}^2 \operatorname{tr}(U^\top (I - P_\star) U) = \underline{\sigma}^2 \|(I - P_\star)U\|_F^2. \tag{201}$$

Combining (199) and (201) gives

$$\|(I - P_\star)U\|_F \leq \underline{\sigma}^{-1} \|X - X_\star\|_F. \tag{202}$$

Finally, direct trace expansion for the two rank-s projectors gives

$$\begin{aligned} \left\|UU^\top - P_\star\right\|_F^2 &= \operatorname{tr}(UU^\top) + \operatorname{tr}(P_\star) - 2\operatorname{tr}(UU^\top P_\star) \\ &= 2s - 2\operatorname{tr}(U^\top P_\star U) = 2\|(I - P_\star)U\|_F^2. \end{aligned} \tag{203}$$

Substitution of (202) proves (197). □

Lemma E.25 (Procrustes alignment from projector closeness)**.** *For every $U, U_\star \in \mathrm{St}(n, s)$, there is $O \in O(s)$ such that $\widehat{U} := UO$ spans the same subspace as U and*

$$\left\|\widehat{U} - U_\star\right\|_2 \leq \left\|\widehat{U} - U_\star\right\|_F \leq \left\|UU^\top - U_\star U_\star^\top\right\|_F. \tag{204}$$

Proof. Write a singular value decomposition in principal-angle form,

$$U^\top U_\star = L \operatorname{diag}(\cos\theta_1, \ldots, \cos\theta_s) R^\top, \qquad 0 \leq \theta_i \leq \frac{\pi}{2}. \tag{205}$$

Take $O = LR^\top$. Direct trace expansions give

$$\|UO - U_\star\|_F^2 = 2\sum_{i=1}^{s}(1 - \cos\theta_i), \tag{206}$$

$$\left\|UU^\top - U_\star U_\star^\top\right\|_F^2 = 2\sum_{i=1}^{s}\sin^2\theta_i. \tag{207}$$

For $0 \leq \theta \leq \pi/2$, $\sin^2\theta = (1 - \cos\theta)(1 + \cos\theta) \geq 1 - \cos\theta$. This proves the Frobenius inequality, and the operator norm is bounded by the Frobenius norm. Finally, $(UO)(UO)^\top = UU^\top$. □

Lemma E.26 (Uniform injectivity on the current active support)**.** *After decreasing r_{trg} if necessary, every target factorization in Corollary E.23 admits an orthogonal change of active basis $\widehat{U} = UO$ such that*

$$\left\|\widehat{U} - U_\star\right\|_2 \leq \frac{\gamma_\star}{4L_A},$$

and

$$\left\|\mathcal{A}(\widehat{U}A\widehat{U}^\top)\right\|_2 \geq \underline{\gamma}\|A\|_F, \qquad \underline{\gamma} := \gamma_\star/2, \qquad A \in \mathbb{S}^s. \tag{208}$$

The coefficient matrix must simultaneously be replaced by $O^\top \Sigma O$, so the represented target is unchanged.

Proof. Let $\bar{P} \in \mathfrak{P}_\star$ be a nearest base projector. The chart construction in the proof of Corollary E.23, specifically (195), places the current parameter and $(\rho, y_\star, \bar{P})$ in one selected chart. Integrating $D\mathcal{X}_j$ along the joining segment gives

$$\left\| X_\rho^\sharp(y,P) - X_\star \right\|_F \le L_{X,1}\big(\|y - y_\star\|_2 + \left\|P - \bar{P}\right\|_F\big),$$

which tends to zero uniformly with r_{trg}. By Lemma E.24, the corresponding support projectors are uniformly close. Lemma E.25 then supplies $O \in O(s)$ with $\|UO - U_\star\|_2$ no larger than the Frobenius distance between those projectors. Shrink r_{trg} to obtain the displayed bound.

For any $A \in \mathbb{S}^s$,

$$\left\| \widehat{U} A \widehat{U}^\top - U_\star A U_\star^\top \right\|_F \le 2 \left\| \widehat{U} - U_\star \right\|_2 \|A\|_F .$$

Hence Lemma E.2 and the definition of L_A give

$$\left\| \mathcal{A}(\widehat{U} A \widehat{U}^\top) \right\|_2 \ge \big(\gamma_\star - 2L_A \left\| \widehat{U} - U_\star \right\|_2\big) \|A\|_F \ge \frac{\gamma_\star}{2} \|A\|_F .$$

□

E.3 Uniform quadratic growth

Theorem E.27 (Uniform reduced quadratic growth)**.** *Use the common target neighborhood and uniform constants in Corollary E.23 and Lemma E.26, and set*

$$\mu_X := \min\left\{ \frac{\underline{\gamma}^2}{2},\ \frac{\underline{\nu}}{1 + 2L_A^2/\underline{\gamma}^2} \right\} > 0. \tag{209}$$

For every parameter triple in Corollary E.23 and every $X \in \mathcal{D}(P)$,

$$\Psi_{\rho,y}(X) - \Psi_{\rho,y}(X_\rho^\sharp(y,P)) \ge \frac{\mu_X}{2} \left\| X - X_\rho^\sharp(y,P) \right\|_F^2 . \tag{210}$$

Thus the estimate is uniform for nearby parameters and holds throughout their respective reduced feasible sets.

Proof. Fix an admissible parameter triple and abbreviate $X^\sharp = X_\rho^\sharp(y,P)$. By Corollary E.23 and Lemma E.26, choose an orthogonal basis $[U, R_c]$ of $\mathrm{Range}(P)$ such that

$$X^\sharp = U \Sigma U^\top, \qquad Z^\sharp = R_c N R_c^\top, \qquad N \succeq \underline{\nu} I, \tag{211}$$

and

$$\left\| \mathcal{A}(U A U^\top) \right\|_2 \ge \underline{\gamma} \|A\|_F \qquad (A \in \mathbb{S}^s). \tag{212}$$

Complementarity and the ranks $(s, r-s)$ justify this orthogonal decomposition.

Step 1: the trace bound controls the nonactive blocks. For any $X \in \mathcal{D}(P)$, write

$$X = [U, R_c] \begin{bmatrix} D & B \\ B^\top & C \end{bmatrix} [U, R_c]^\top, \qquad A := D - \Sigma, \qquad E := X - X^\sharp. \tag{213}$$

Set $t := \operatorname{tr} C$ and $E_\perp := E - UAU^\top$. PSD feasibility gives $D, C \succeq 0$, while $\operatorname{tr} X = \operatorname{tr} D + t \le 1$. Each two-by-two principal minor gives $B_{ij}^2 \le D_{ii} C_{jj}$; summing these inequalities and using $\|C\|_F^2 \le (\operatorname{tr} C)^2$ yields

$$0 \le t \le 1, \qquad \|B\|_F^2 \le (\operatorname{tr} D) t \le (1-t)t, \qquad \|C\|_F^2 \le t^2. \tag{214}$$

The blocks are Frobenius orthogonal, so

$$\|E_\perp\|_F^2 = 2\|B\|_F^2 + \|C\|_F^2 \le 2t - t^2 \le 2t, \qquad \|E\|_F^2 = \|A\|_F^2 + \|E_\perp\|_F^2. \tag{215}$$

Step 2: active-block injectivity controls the remaining error. Let $a := \|\mathcal{A}(E)\|_2$. Since $UAU^\top = E - E_\perp$,

$$\underline{\gamma}\,\|A\|_F \le a + L_A\,\|E_\perp\|_F. \tag{216}$$

Squaring and using (215) gives

$$\|E\|_F^2 \le \frac{2}{\underline{\gamma}^2}a^2 + 2\left(1 + \frac{2L_A^2}{\underline{\gamma}^2}\right)t. \tag{217}$$

Step 3: the objective controls both quantities. The quadratic objective has the exact expansion

$$\Psi_{\rho,y}(X) - \Psi_{\rho,y}(X^\sharp) = \left\langle \nabla\Psi_{\rho,y}(X^\sharp), E\right\rangle + \frac{1}{2}\|\mathcal{A}(E)\|_2^2. \tag{218}$$

At the trace-inactive target, $P\nabla\Psi_{\rho,y}(X^\sharp)P = Z^\sharp$. Because $E = PEP$, the compressed stationarity equation gives

$$\left\langle \nabla\Psi_{\rho,y}(X^\sharp), E\right\rangle = \left\langle P\nabla\Psi_{\rho,y}(X^\sharp)P, E\right\rangle = \left\langle Z^\sharp, E\right\rangle = \langle N, C\rangle. \tag{219}$$

Consequently,

$$\Psi_{\rho,y}(X) - \Psi_{\rho,y}(X^\sharp) = \frac{1}{2}a^2 + \langle N, C\rangle \ge \frac{1}{2}a^2 + \underline{\nu}t. \tag{220}$$

The choice (209) ensures $\mu_X/\underline{\gamma}^2 \le 1/2$ and $\mu_X(1 + 2L_A^2/\underline{\gamma}^2) \le \underline{\nu}$. Multiplying (217) by $\mu_X/2$ and comparing with (220) proves (210). All constants are uniform over the prescribed parameter neighborhood. □

E.4 Tracking by the Ritz gap

Lemma E.28 (The dual displacement and Ritz gap control leakage of $U_\star$)**.** *Let $V \in \mathrm{St}(n,r)$, $P = VV^\top$, and $y \in \mathcal{Y}$. Then*

$$\begin{aligned}\|(I-P)U_\star\|_F &\le L_P\,\|y - y_\star\|_2 + \sqrt{\frac{\Phi_s(H(y),V)}{\underline{\Delta}}} \\ &\le C_U\left(\|y - y_\star\|_2 + \sqrt{\Phi_s(H(y),V)}\right), \qquad C_U := \max\left\{L_P, \underline{\Delta}^{-1/2}\right\}.\end{aligned} \tag{221}$$

Proof. Let U_y span the top-s eigenspace of $H(y)$ and $P_y = U_yU_y^\top$. If Q is the best s-dimensional Ritz subspace in $\mathrm{Range}(V)$, expansion in an eigenbasis of $H(y)$ gives

$$\Phi_s(H(y),V) \ge \underline{\Delta}\,\|(I-P_y)Q\|_F^2.$$

Because $QQ^\top \preceq P$, the equal-rank projector identity yields

$$\|(I-P)U_y\|_F^2 \le \|(I-P_y)Q\|_F^2 \le \frac{\Phi_s(H(y),V)}{\underline{\Delta}}.$$

Now insert P_y:

$$(I-P)U_\star = (I-P)(P_\star - P_y)U_\star + (I-P)U_y(U_y^\top U_\star).$$

The first term is at most $\|P_y - P_\star\|_F$, while the second is at most $\|(I-P)U_y\|_F$ because $\left\|U_y^\top U_\star\right\|_2 \le 1$. Apply Lemma 5.2. □

Lemma E.29 (A nearby redundant base projector)**.** *If $P \in \mathfrak{P}_r$ and $\|(I-P)U_\star\|_2 < 1$, there is an explicitly constructed $\bar{P} \in \mathfrak{P}_\star$ satisfying*

$$\left\|P - \bar{P}\right\|_F = \sqrt{2}\,\|(I-P)U_\star\|_F\,. \tag{222}$$

Proof. Set $G = U_\star^\top P U_\star$. For $a \in \mathbb{R}^s$,

$$a^\top G a = \|PU_\star a\|_2^2 \geq (1 - \|(I-P)U_\star\|_2^2)\,\|a\|_2^2\,,$$

so $G \succ 0$. Define

$$Q := PU_\star G^{-1/2} \in \mathrm{St}(n,s)$$

and complete it within $\mathrm{Range}(P)$ to $[Q, R_P] \in \mathrm{St}(n,r)$. Since $R_P^\top U_\star = 0$, the projector

$$\bar{P} := P_\star + R_P R_P^\top \in \mathfrak{P}_\star \tag{223}$$

is well defined. Moreover, $\mathrm{Range}(Q) = \mathrm{Range}(PU_\star)$ and the orthogonal decomposition $U_\star = PU_\star + (I-P)U_\star$ shows $QQ^\top U_\star = PU_\star$. Since $P - \bar{P} = QQ^\top - P_\star$, the equal-rank projector identity gives

$$\left\|P - \bar{P}\right\|_F^2 = 2s - 2\,\mathrm{tr}(U_\star^\top P U_\star) = 2\,\|(I-P)U_\star\|_F^2\,.$$

□

Corollary E.30 (Ritz error controls distance to the redundant projector family)**.** *Let $V \in \mathrm{St}(n,r)$, $P = VV^\top$, $y \in \mathcal{Y}$, and suppose*

$$C_U\left(\|y - y_\star\|_2 + \sqrt{\Phi_s(H(y),V)}\right) < 1. \tag{224}$$

Then

$$\mathrm{dist}_F(P, \mathfrak{P}_\star) \leq C_P\left(\|y - y_\star\|_2 + \sqrt{\Phi_s(H(y),V)}\right), \quad C_P := \max\left\{\sqrt{2}L_P, \sqrt{2/\underline{\Delta}}\right\}. \tag{225}$$

Proof. Lemma E.28 and (224) first give $\|(I-P)U_\star\|_2 < 1$. Hence Lemma E.29 constructs $\bar{P} \in \mathfrak{P}_\star$ and gives

$$\begin{aligned}\left\|P - \bar{P}\right\|_F &= \sqrt{2}\,\|(I-P)U_\star\|_F \\ &\leq \sqrt{2}L_P\,\|y - y_\star\|_2 + \sqrt{\frac{2}{\underline{\Delta}}}\sqrt{\Phi_s(H(y),V)} \\ &\leq C_P\left(\|y - y_\star\|_2 + \sqrt{\Phi_s(H(y),V)}\right).\end{aligned}$$

Taking the infimum over $\mathfrak{P}_\star$ can only decrease the left side and proves (225). □

Theorem E.31 (Quantitative tracking of the exact target)**.** *Let*

$$r_{\mathrm{track}} := \min\left\{R_Y, \frac{1}{2C_U}, \frac{\delta_{\mathrm{chart}}}{4(1+C_P)}\right\}. \tag{226}$$

If $\rho \in [\rho_-, \rho_+]$, $V \in \mathrm{St}(n,r)$, $P = VV^\top$, and

$$\varepsilon(y,V) := \|y - y_\star\|_2 + \sqrt{\Phi_s(H(y),V)} \leq r_{\mathrm{track}}, \tag{227}$$

then the exact reduced target is well defined and satisfies

$$\left\|X_\rho^\sharp(y,P) - X_\star\right\|_F \leq L_{X,1}(1+C_P)\left(\|y - y_\star\|_2 + \sqrt{\Phi_s(H(y),V)}\right). \tag{228}$$

Proof. Write $\varepsilon = \|y - y_\star\|_2 + \sqrt{\Phi_s(H(y), V)}$. First, $\varepsilon \le R_Y$ places y in the single spectral ball used throughout the paper. Next,

$$C_U \varepsilon \le \frac{1}{2} < 1. \tag{229}$$

Thus Lemma E.28 proves the angle premise of Lemma E.29, which in turn constructs a specific $\bar{P} \in \mathfrak{P}_\star$. The two lemmas and (225) give

$$\left\|P - \bar{P}\right\|_F \le C_P \varepsilon. \tag{230}$$

Let $a = (\rho, y_\star, \bar{P})$ and $\theta = (\rho, y, P)$. Then

$$\|\theta - a\| \le \|y - y_\star\|_2 + \left\|P - \bar{P}\right\|_F \le (1 + C_P)\varepsilon \le \frac{\delta_{\text{chart}}}{4}. \tag{231}$$

Because $a \in \mathcal{K}_0$, choose a selected chart j with $a \in B_{r_j}(a_j)$. Since $\delta_{\text{chart}} \le r_j$,

$$\|\theta - a_j\| \le \|\theta - a\| + \|a - a_j\| < \frac{r_j}{4} + r_j < 2r_j. \tag{232}$$

Thus both endpoints and the segment $\theta(t) = a + t(\theta - a)$ lie in the convex set $B_{2r_j}(a_j) \subset K_j \subset \mathcal{U}_j$.

Both endpoint third arguments are actual projectors. Hence the branch identification in Theorem E.17 gives

$$\mathcal{X}_j(\theta) = X_\rho^\sharp(y, P), \qquad \mathcal{X}_j(a) = X_\rho^\sharp(y_\star, \bar{P}) = X_\star, \tag{233}$$

where the last equality is Lemma E.4. The fundamental theorem of calculus on this fixed chart now yields

$$\begin{aligned}\left\|X_\rho^\sharp(y, P) - X_\star\right\|_F &\le \int_0^1 \|D\mathcal{X}_j(\theta(t))\|_{\text{op}} \, \|\theta - a\| \, dt \\ &\le L_{X,1} \, \|\theta - a\| \le L_{X,1}(1 + C_P)\varepsilon,\end{aligned}$$

which is (228). The same branch theorem also proves that this point is the unique global minimizer of the actual reduced problem. □

E.5 Product drift under one residual step

Lemma E.32 (Two exact-zero identities for the drift map)**.** *For every selected chart j and every point in its natural mismatch domain, the following hold:*

(i) if $P \in \mathfrak{P}_r$, then

$$\mathcal{G}_j(\rho, y, P, 0) = 0. \tag{234}$$

(ii) if $P \in \mathfrak{P}_\star$, then

$$\mathcal{G}_j(\rho, y_\star, P, \eta) = 0, \qquad \partial_\eta \mathcal{G}_j(\rho, y_\star, P, \eta) = 0. \tag{235}$$

Proof. For (i), the actual-projector identity for the ambient update gives $\mathcal{P}_j(y, P, 0) = P^+(y, P, 0) = P$. Both target terms in (182) are therefore the same matrix $X_\rho^\sharp(y, P)$, and this matrix satisfies $PX_\rho^\sharp(y, P)P = X_\rho^\sharp(y, P)$. Hence the mismatch is zero.

For (ii), Lemma E.19 gives $P_\eta := P^+(y_\star, P, \eta) \in \mathfrak{P}_\star$. Both triples occur in the natural mismatch domain, so the branch maps coincide with the actual targets. By Lemma E.4,

$$X_\rho^\sharp(y_\star, P) = X_\star, \qquad X_\rho^\sharp(y_\star, P_\eta) = X_\star. \tag{236}$$

Moreover, $P_\eta \in \mathfrak{P}_\star$ contains $\mathrm{Range}(X_\star)$ and hence $P_\eta X_\star P_\eta = X_\star$. Substitution in (182) proves the first identity in (235).

Finally, the mismatch domain is open. Therefore the same zero identity holds for every step $\eta + h$ in a sufficiently small interval about η. Taking the difference quotient gives $\partial_\eta \mathcal{G}_j(\rho, y_\star, P, \eta) = 0$. □

Theorem E.33 (Product-form target drift after one full-residual step)**.** *There exist $C_T < \infty$, $r_{\mathrm{drift}} > 0$, and $\bar{\eta}_{\mathrm{drift}} > 0$ such that, whenever*

$$\rho \in [\rho_-, \rho_+], \qquad \|y - y_\star\|_2 \le r_{\mathrm{drift}}, \qquad V \in \mathrm{St}(n, r), \quad P = VV^\top,$$
$$\Phi_s(H(y), V) \le r_{\mathrm{drift}}^2, \qquad 0 \le \eta \le \bar{\eta}_{\mathrm{drift}},$$

the two exact targets are well defined, and the updated projector $P^+ = P^+(y, P, \eta)$ satisfies

$$\left\| P^+ X_\rho^\sharp(y, P) P^+ - X_\rho^\sharp(y, P^+) \right\|_F^2 \le C_T \eta^2 \left(\|y - y_\star\|_2^2 + \Phi_s(H(y), V) \right). \tag{237}$$

Proof. Set $\varepsilon = \|y - y_\star\|_2 + \sqrt{\Phi_s(H(y), V)}$ and choose

$$r_{\mathrm{drift}} \le \min\left\{ R_Y, \frac{1}{4C_U}, \frac{\delta_{\mathrm{chart}}}{4(1 + C_P)} \right\}, \qquad \bar{\eta}_{\mathrm{drift}} \le \eta_{\mathrm{chart}}. \tag{238}$$

The theorem hypotheses imply $\varepsilon \le 2r_{\mathrm{drift}}$.

Step 1: construct a corresponding base projector. Since $y \in \mathcal{Y}$ and $C_U \varepsilon \le 2C_U r_{\mathrm{drift}} \le 1/2$, the leakage lemma and then the nearby-projector lemma apply in this order. They produce $\bar{P} \in \mathfrak{P}_\star$ satisfying

$$\left\| P - \bar{P} \right\|_F \le C_P \varepsilon. \tag{239}$$

With $\theta = (\rho, y, P)$ and $a = (\rho, y_\star, \bar{P})$,

$$\|\theta - a\| \le (1 + C_P)\varepsilon \le \frac{\delta_{\mathrm{chart}}}{2}. \tag{240}$$

Step 2: keep both parameter paths in one selected compact box. Choose j with $a \in B_{r_j}(a_j)$. Because $\delta_{\mathrm{chart}} \le r_j$, (240) gives

$$a, \theta \in B_{2r_j}(a_j) \subset \overline{B}_{4r_j}(a_j) = K_j. \tag{241}$$

For every $t \in [0, \eta]$, $0 \le t \le \bar{\eta}_{\mathrm{drift}} \le \eta_{\mathrm{chart}} \le \eta_j$; hence

$$(\theta, t), (a, t) \in K_j^G = K_j \times [-\eta_j, \eta_j] \subset \mathcal{D}_j^G. \tag{242}$$

This membership verifies simultaneously that both branch targets and the updated projector inside the second target are defined. It also places, for each fixed t, the entire segment between (θ, t) and (a, t) inside the convex work box.

Step 3: compare the current step derivative with the exact-zero base derivative. The definition of L_G and the mean-value inequality give

$$\left\| \partial_\eta \mathcal{G}_j(\rho, y, P, t) - \partial_\eta \mathcal{G}_j(\rho, y_\star, \bar{P}, t) \right\|_F \le L_G \left\| \theta - a \right\|. \tag{243}$$

By Lemma E.32(ii), the second derivative term is exactly zero. Thus, uniformly for $t \in [0, \eta]$,

$$\left\| \partial_\eta \mathcal{G}_j(\rho, y, P, t) \right\|_F \le L_G (1 + C_P)\varepsilon. \tag{244}$$

Step 4: integrate only after the full path has been validated. Equation (242) at $t = 0$ permits Lemma E.32(i), so $\mathcal{G}_j(\rho, y, P, 0) = 0$. The fundamental theorem of calculus and (244) yield

$$\begin{aligned}\|\mathcal{G}_j(\rho, y, P, \eta)\|_F &\le \int_0^\eta \|\partial_t \mathcal{G}_j(\rho, y, P, t)\|_F \, dt \\ &\le L_G(1 + C_P)\eta\varepsilon. \end{aligned} \tag{245}$$

On actual projectors, the chart update equals $P^+(y, P, \eta)$ and both branch targets equal the corresponding unique actual reduced targets. Therefore (182) is precisely the matrix on the left of (237). Squaring (245) and using $(a + b)^2 \le 2a^2 + 2b^2$ proves (237) with

$$C_T := 2[L_G(1 + C_P)]^2. \tag{246}$$

□

The one-X step uses the following standard consequence of smoothness and quadratic growth. We include its short proof to fix the contraction factor.

Lemma E.34 (One projected-gradient step under quadratic growth)**.** *Let $\mathcal{C}$ be a nonempty closed convex subset of a finite-dimensional Euclidean space. Let f be differentiable on a neighborhood of $\mathcal{C}$, convex and L-smooth on $\mathcal{C}$, with $L > 0$ and unique minimizer $x^\star$. Suppose that, for some $\mu > 0$,*

$$f(z) - f(x^\star) \ge \frac{\mu}{2} \|z - x^\star\|^2 \qquad (z \in \mathcal{C}).$$

For every $x \in \mathcal{C}$ and $0 < \eta \le 1/L$, the update $x^+ = \mathrm{Proj}_{\mathcal{C}}(x - \eta \nabla f(x))$ satisfies

$$\left\|x^+ - x^\star\right\|^2 \le \frac{1}{1 + \mu\eta} \|x - x^\star\|^2 . \tag{247}$$

Proof. The projection variational inequality is

$$\left\langle x^+ - x + \eta \nabla f(x), z - x^+ \right\rangle \ge 0 \qquad (z \in \mathcal{C}). \tag{248}$$

Taking $z = x^\star$ and rearranging gives

$$\left\langle \nabla f(x), x^+ - x^\star \right\rangle \le \frac{1}{\eta} \left\langle x - x^+, x^+ - x^\star \right\rangle . \tag{249}$$

Convexity at x and the descent lemma from x to x^+ yield

$$\begin{aligned} f(x^+) - f(x^\star) &\le \langle \nabla f(x), x - x^\star \rangle + \left\langle \nabla f(x), x^+ - x \right\rangle + \frac{L}{2} \left\|x^+ - x\right\|^2 \\ &= \left\langle \nabla f(x), x^+ - x^\star \right\rangle + \frac{L}{2} \left\|x^+ - x\right\|^2 . \end{aligned} \tag{250}$$

Substitute (249) and use the three-point identity

$$2 \left\langle x - x^+, x^+ - x^\star \right\rangle = \|x - x^\star\|^2 - \left\|x^+ - x^\star\right\|^2 - \left\|x - x^+\right\|^2 . \tag{251}$$

This gives

$$f(x^+) - f(x^\star) \le \frac{1}{2\eta} [\|x - x^\star\|^2 - \left\|x^+ - x^\star\right\|^2]$$

$$-\left(\frac{1}{2\eta}-\frac{L}{2}\right)\left\|x-x^{+}\right\|^{2}. \tag{252}$$

Since $x^{+}\in\mathcal{C}$, quadratic growth applies at x^{+}. Moreover, $\eta\le 1/L$ makes the last coefficient in (252) nonnegative. Therefore

$$\frac{\mu}{2}\left\|x^{+}-x^{\star}\right\|^{2}\le\frac{1}{2\eta}[\|x-x^{\star}\|^{2}-\left\|x^{+}-x^{\star}\right\|^{2}]. \tag{253}$$

Multiplication by 2η and rearrangement prove (247). □

Proof of Corollary 5.12. Equation (20) is L_A^2-Lipschitz in X. By Theorem E.27, quadratic growth holds for every matrix in $\mathcal{D}(P)$. Apply Lemma E.34 on this closed convex set with $L=L_A^2$ and $\mu=\mu_X$. □

F Proofs for Section 5.4

Assumptions 3.3 and 5.7 are in force throughout this appendix, as in Section 5.4. The proofs proceed from exact proximal control and the three quantitative recurrences to localization, explicit parameter feasibility, and induction with the weighted scalar error.

Proof of Lemma 5.14. The function $z\mapsto M_{V_k}(z)+(\rho/2)\|z-y_k\|^2$ is ρ-strongly convex. Its minimizer $\widehat{y}_{k+1}$ therefore satisfies, for every $z\in\mathbb{R}^m$,

$$\begin{aligned}M_{V_k}(\widehat{y}_{k+1})+\frac{\rho}{2}\|\widehat{y}_{k+1}-y_k\|_2^2+\frac{\rho}{2}\|\widehat{y}_{k+1}-z\|_2^2\\ \le M_{V_k}(z)+\frac{\rho}{2}\|z-y_k\|_2^2.\end{aligned} \tag{254}$$

Set $z=y_\star$. Global minorization gives $M_{V_k}(y_\star)\le F(y_\star)$, whereas Lemma 5.9, with center y_k and evaluation point $\widehat{y}_{k+1}$, gives

$$M_{V_k}(\widehat{y}_{k+1})\ge F(\widehat{y}_{k+1})-\frac{\beta}{2}\|\widehat{y}_{k+1}-y_k\|_2^2-c_\Phi\sigma_k. \tag{255}$$

Substitution into (254) yields

$$\begin{aligned}F(\widehat{y}_{k+1})-F(y_\star)+\frac{\rho-\beta}{2}\|\widehat{y}_{k+1}-y_k\|_2^2\\ +\frac{\rho}{2}\|\widehat{y}_{k+1}-y_\star\|_2^2\le\frac{\rho}{2}d_k+c_\Phi\sigma_k.\end{aligned} \tag{256}$$

By (56) and Assumption 5.7,

$$F(\widehat{y}_{k+1})-F(y_\star)\ge\frac{\mu_y}{2}\|\widehat{y}_{k+1}-y_\star\|_2^2.$$

Insert this lower bound into (256) and discard the nonnegative displacement term to obtain (57). Alternatively, discard the objective gap and target-distance terms. Since $\rho>\beta$ by (52), the result is (58). □

Proof of Lemma 5.15. Let $\widetilde{y}_{k+1} = y_k - \rho^{-1}(\mathcal{A}(X_k) - b)$. Then

$$\|\widetilde{y}_{k+1} - \widehat{y}_{k+1}\|_2^2 \le \frac{L_A^2}{\rho^2}\xi_k. \tag{257}$$

Indeed, the difference equals $-\rho^{-1}\mathcal{A}(X_k - X_k^\sharp)$, so the definition of L_A proves (257).

Set $\epsilon = \mu_y/(2\rho)$. Young's inequality and Lemma 5.14, which is applicable precisely under (60)–(61), yield

$$\begin{aligned}\|\widetilde{y}_{k+1} - y_\star\|_2^2 &\le (1+\epsilon)\,\|\widehat{y}_{k+1} - y_\star\|_2^2 + (1+\epsilon^{-1})\,\|\widetilde{y}_{k+1} - \widehat{y}_{k+1}\|_2^2 \\ &\le \left(1 - \frac{\mu_y}{2(\rho+\mu_y)}\right) d_k \\ &\quad + \left(1 + \frac{\mu_y}{2\rho}\right)\frac{2c_\Phi}{\rho+\mu_y}\sigma_k + \left(1 + \frac{2\rho}{\mu_y}\right)\frac{L_A^2}{\rho^2}\xi_k.\end{aligned} \tag{258}$$

Since $y_{k+1} = (1-\tau)y_k + \tau\widetilde{y}_{k+1}$, convexity of the squared norm gives

$$d_{k+1} \le (1-\tau)d_k + \tau\,\|\widetilde{y}_{k+1} - y_\star\|_2^2\,. \tag{259}$$

Substitution of (258) proves (62) coefficient by coefficient.

For the displacement, use the exact decomposition

$$y_{k+1} - y_k = \tau\big[(\widehat{y}_{k+1} - y_k) + (\widetilde{y}_{k+1} - \widehat{y}_{k+1})\big]. \tag{260}$$

Applying $\|a+b\|^2 \le 2\,\|a\|^2 + 2\,\|b\|^2$, then (58) and (257), gives

$$\|y_{k+1} - y_k\|_2^2 \le \tau^2\left[\frac{2\rho}{\rho-\beta}d_k + \frac{4c_\Phi}{\rho-\beta}\sigma_k + \frac{2L_A^2}{\rho^2}\xi_k\right].$$

This is (63) by (59). □

Proof of Lemma 5.16. The threshold-free part follows first. Since both dual points lie in the same convex spectral ball, (33) gives

$$\widetilde{\sigma}_{k+1} \le \sigma_k + C_1\sqrt{\sigma_k}\,\|y_{k+1} - y_k\|_2 + C_0\,\|y_{k+1} - y_k\|_2^2\,. \tag{261}$$

Young's inequality applied to the cross term gives

$$C_1\sqrt{\sigma_k}\,\|y_{k+1} - y_k\| \le \frac{\kappa_V\eta_V}{4}\sigma_k + \frac{C_1^2}{\kappa_V\eta_V}\,\|y_{k+1} - y_k\|^2\,.$$

Combining this inequality with (261) and the definition of g_V proves (65). No one-V threshold has entered this argument.

Only after (66) has been established independently may Theorem 5.6 be invoked for the fixed matrix $H(y_{k+1})$; it gives

$$\sigma_{k+1} \le q_V\widetilde{\sigma}_{k+1}. \tag{262}$$

The explicit premises of Lemma 5.15 make (63) available. Substituting that displacement bound into (65), and then into (262), proves (67). □

Proof of Lemma 5.17. Step 1: motion of the target with y at fixed P_k. Choose $\bar{P} \in \mathfrak{P}_\star$ nearest to P_k and put $a = (\rho, y_\star, \bar{P})$. The conditions in (73) place both (ρ, y_k, P_k) and (ρ, y_{k+1}, P_k) at distance less than $2r_{\mathrm{trg}} < \delta_{\mathrm{chart}}$ from a. Select the chart index using $a \in B_{r_j}(a_j)$, as in the proof of Corollary E.23. The endpoint estimate (194) puts both parameters in $B_{2r_j}(a_j)$, so their joining segment remains in the same compact work set K_j. Integrating its target branch along this segment gives

$$\left\| X_\rho^\sharp(y_k, P_k) - X_\rho^\sharp(y_{k+1}, P_k) \right\|_F \le L_{X,1} \left\| y_{k+1} - y_k \right\|_2 . \tag{263}$$

Step 2: motion of the target with the projector. Split (68) exactly as

$$\begin{aligned} R_k = {} & P_{k+1}[X_\rho^\sharp(y_k, P_k) - X_\rho^\sharp(y_{k+1}, P_k)]P_{k+1} \\ & + [P_{k+1}X_\rho^\sharp(y_{k+1}, P_k)P_{k+1} - X_\rho^\sharp(y_{k+1}, P_{k+1})]. \end{aligned}$$

The actual algorithmic update satisfies $P_{k+1} = P^+(y_{k+1}, P_k, \eta_V)$. Hence the three explicit conditions in (74) make Theorem E.33 applicable, and it yields

$$\begin{aligned} & \left\| P_{k+1}X_\rho^\sharp(y_{k+1}, P_k)P_{k+1} - X_\rho^\sharp(y_{k+1}, P_{k+1}) \right\|_F^2 \\ & \qquad \le C_T \eta_V^2 \big(d_{k+1} + \widetilde{\sigma}_{k+1} \big). \end{aligned} \tag{264}$$

Combining (263) and (264) with $\|A + B\|_F^2 \le 2\|A\|_F^2 + 2\|B\|_F^2$ gives

$$\|R_k\|_F^2 \le 2L_{X,1}^2 \|y_{k+1} - y_k\|_2^2 + 2C_T\eta_V^2(d_{k+1} + \widetilde{\sigma}_{k+1}). \tag{265}$$

The elementary relation

$$d_{k+1} = \|(y_k - y_\star) + (y_{k+1} - y_k)\|_2^2 \le 2d_k + 2\|y_{k+1} - y_k\|_2^2 \tag{266}$$

then transforms (265) into

$$\begin{aligned} \|R_k\|_F^2 \le {} & (2L_{X,1}^2 + 4C_T\eta_V^2) \|y_{k+1} - y_k\|_2^2 \\ & + 4C_T\eta_V^2 d_k + 2C_T\eta_V^2 \widetilde{\sigma}_{k+1}. \end{aligned} \tag{267}$$

Because all explicit premises of Lemma 5.15 and the threshold-free part of Lemma 5.16 were assumed, we may now substitute

$$\|y_{k+1} - y_k\|_2^2 \le \tau^2(a_d d_k + a_\sigma \sigma_k + a_\xi \xi_k), \tag{268}$$

$$\widetilde{\sigma}_{k+1} \le \left(1 + \frac{\kappa_V \eta_V}{4}\right) \sigma_k + g_V \tau^2 (a_d d_k + a_\sigma \sigma_k + a_\xi \xi_k). \tag{269}$$

Insertion into (267), followed by collecting the coefficients can be displayed without any hidden estimate. Put

$$A_T := 2L_{X,1}^2 + 4C_T\eta_V^2, \qquad \mathcal{B}_k := a_d d_k + a_\sigma \sigma_k + a_\xi \xi_k. \tag{270}$$

Then (267) becomes

$$\begin{aligned} \|R_k\|_F^2 \le {} & A_T \tau^2 \mathcal{B}_k + 4C_T\eta_V^2 d_k \\ & + 2C_T\eta_V^2 \left[\left(1 + \frac{\kappa_V \eta_V}{4}\right) \sigma_k + g_V \tau^2 \mathcal{B}_k \right]. \end{aligned} \tag{271}$$

Consequently, the coefficients of the three old errors are respectively

$$[d_k]: \quad A_T\tau^2 a_d + 4C_T\eta_V^2 + 2C_T\eta_V^2 g_V \tau^2 a_d,$$

$$
\begin{aligned}
[\sigma_k]: \quad & A_T\tau^2 a_\sigma + 2C_T\eta_V^2\left(1+\frac{\kappa_V\eta_V}{4}+g_V\tau^2 a_\sigma\right),\\
[\xi_k]: \quad & A_T\tau^2 a_\xi + 2C_T\eta_V^2 g_V\tau^2 a_\xi.
\end{aligned}
\tag{272}
$$

These are exactly r_d, r_σ, r_ξ in (71), so

$$
\|R_k\|_F^2 \le r_d d_k + r_\sigma\sigma_k + r_\xi\xi_k \tag{273}
$$

Step 3: control the predictor error. By Frobenius nonexpansiveness of two-sided orthogonal compression,

$$
\left\|P_{k+1}(X_k - X_k^\sharp)P_{k+1}\right\|_F \le \sqrt{\xi_k}. \tag{274}
$$

Applying Young's inequality with the positive parameter $\mu_X\eta_X/2$ to the two terms in (69) gives

$$
\begin{aligned}
\left\|\widehat{X}_{k+1} - X_{k+1}^\sharp\right\|_F^2 \le\ & \left(1+\frac{\mu_X\eta_X}{2}\right)\xi_k\\
& + \left(1+\frac{2}{\mu_X\eta_X}\right)\|R_k\|_F^2.
\end{aligned}
\tag{275}
$$

Step 4: contract toward the new target. Conditions (73) and (75) place the new target in the parameter neighborhood of Theorem E.27. The predictor $\widehat{X}_{k+1} = P_{k+1}X_kP_{k+1}$ belongs to $\mathcal{D}(P_{k+1})$ by Lemma 4.1, and the step bound $0 < \eta_X \le L_A^{-2}$ is part of (75). Thus Corollary 5.12 gives

$$
\xi_{k+1} \le \frac{1}{1+\mu_X\eta_X}\left\|\widehat{X}_{k+1} - X_{k+1}^\sharp\right\|_F^2. \tag{276}
$$

Substituting (275) and then (273) into (276), and using (70), gives (76). □

Proof of Theorem 5.18. We verify the local premises in a one-way order.

Step 1: construct the current target before defining its primal error. Put

$$
\varepsilon_k := \sqrt{d_k} + \sqrt{\sigma_k}. \tag{277}
$$

By the first term in (82) and the extra factor $1/2$ in its definition,

$$
\varepsilon_k \le 2\sqrt{R_0} < r_{\mathrm{track}}. \tag{278}
$$

Therefore Theorem E.31 first proves the existence and uniqueness of $X_k^\sharp$; only now is ξ_k defined. Moreover, (225) and the second term in (82) give

$$
\mathrm{dist}_F(P_k, \mathfrak{P}_\star) \le C_P\varepsilon_k \le 2C_P\sqrt{R_0} < r_{\mathrm{trg}}. \tag{279}
$$

The direct radius terms in (82) also give

$$
\|y_k - y_\star\|_2 = \sqrt{d_k} < \min\{R_Y, r_{\mathrm{trg}}\}. \tag{280}
$$

Step 2: verify every premise of the dual recursion. The global model-prox displacement bound (39) gives $\|\widehat{y}_{k+1} - y_k\|_2 \le G_M/\rho$. By (52), this is at most $\frac{1}{2}\min\{R_Y, r_F\}$. The fifth entry of (82) gives the same upper bound for $\|y_k - y_\star\|_2$. Consequently,

$$
\|\widehat{y}_{k+1} - y_\star\|_2 \le \|y_k - y_\star\|_2 + \frac{G_M}{\rho} \le \min\{R_Y, r_F\}. \tag{281}
$$

Equations (279)– (281) verify explicitly (60)–(61). Hence Lemma 5.15 gives the first row of (85) and

$$\|y_{k+1} - y_k\|_2^2 \le \tau^2 B_{\Delta y} R_0. \tag{282}$$

The triangle inequality then yields

$$\|y_{k+1} - y_\star\|_2 \le (1 + \tau\sqrt{B_{\Delta y}})\sqrt{R_0} < \min\{R_Y, r_{\mathrm{trg}}, r_{\mathrm{drift}}\}. \tag{283}$$

Step 3: obtain a prior moving-H bound, verify the one-V threshold, and only then refresh. The threshold-free estimate (65) and (282) give

$$\widetilde{\sigma}_{k+1} \le B_{\mathrm{pre}} R_0 < \min\left\{\frac{\underline{\Delta}^2}{2\overline{\Gamma}}, r_{\mathrm{drift}}^2\right\}. \tag{284}$$

This proves, rather than assumes, the first inequality in (66); the step-size inequality is part of (77). Thus the second part of Lemma 5.16 is now valid and gives the second row.

The first two matrix rows imply

$$d_{k+1} \le S_1 R_0, \qquad \sigma_{k+1} \le S_2 R_0. \tag{285}$$

Therefore

$$\sqrt{d_{k+1}} + \sqrt{\sigma_{k+1}} \le (\sqrt{S_1} + \sqrt{S_2})\sqrt{R_0} < \min\{r_{\mathrm{track}}, r_{\mathrm{trg}}/C_P\}. \tag{286}$$

Another application of Theorem E.31 proves that the new target $X_{k+1}^\sharp$ is uniquely defined, while (225) gives

$$\mathrm{dist}_F(P_{k+1}, \mathfrak{P}_\star) < r_{\mathrm{trg}}. \tag{287}$$

Step 4: apply the primal recurrence at the new reduced target. The algorithmic state at $k+1$ is feasible, and Step 3 has already constructed its unique target $X_{k+1}^\sharp$. Thus d_{k+1}, σ_{k+1}, and $\xi_{k+1} = \left\|X_{k+1} - X_{k+1}^\sharp\right\|_F^2$ are all well defined. Equations (280), (283), (279), (284), and (287), together with (77), verify every premise (72)–(75) of Lemma 5.17. That lemma controls target drift and the feasible predictor, then uses quadratic growth on $\mathcal{D}(P_{k+1})$ to obtain (76). This is the third row of (85). □

Proof of Theorem 5.19. Fix ρ and η_X as in the statement. Every asymptotic constant below is independent of ε. Set

$$c_1 := \frac{\mu_y}{2(\rho + \mu_y)} > 0, \qquad c_2 := \frac{3\kappa_V}{4} > 0. \tag{288}$$

The constants q_W, h_W in (70) are fixed, and

$$1 - q_W = \frac{\mu_X \eta_X}{2(1 + \mu_X \eta_X)} > 0. \tag{289}$$

Step 1: record the contraction and coupling orders. Since

$$g_V = C_0 + \frac{C_1^2}{\kappa_V \varepsilon} = O(\varepsilon^{-1}), \qquad q_V = 1 - \kappa_V \varepsilon,$$

the first two rows in (78)–(79) satisfy

$$\begin{aligned} M_{11} &= 1 - c_1\varepsilon^2, & M_{12}, M_{13} &= O(\varepsilon^2), \\ M_{21}, M_{23} &= O(\varepsilon^3), & M_{22} &= 1 - c_2\varepsilon + O(\varepsilon^2). \end{aligned} \tag{290}$$

For example,

$$M_{22} = (1 - \kappa_V \varepsilon)\left(1 + \frac{\kappa_V}{4}\varepsilon + g_V a_\sigma \varepsilon^4\right) = 1 - c_2\varepsilon + O(\varepsilon^2).$$

The target-drift coefficients in (71) give

$$r_d = 4C_T\varepsilon^2 + O(\varepsilon^4), \quad r_\sigma = 2C_T\varepsilon^2 + O(\varepsilon^3), \quad r_\xi = O(\varepsilon^4). \tag{291}$$

Consequently,

$$M_{31}, M_{32} = O(\varepsilon^2), \qquad M_{33} = q_W + O(\varepsilon^4). \tag{292}$$

Step 2: estimate the three weighted column sums. With $w = w_\varepsilon$, equations (290)–(292) give

$$\frac{(w^\top M)_1}{w_1} = M_{11} + \sqrt{\varepsilon}\, M_{21} + \sqrt{\varepsilon}\, M_{31} = 1 - c_1\varepsilon^2 + O(\varepsilon^{5/2}), \tag{293}$$

$$\frac{(w^\top M)_2}{w_2} = \varepsilon^{-1/2} M_{12} + M_{22} + M_{32} = 1 - c_2\varepsilon + O(\varepsilon^{3/2}), \tag{294}$$

$$\frac{(w^\top M)_3}{w_3} = \varepsilon^{-1/2} M_{13} + M_{23} + M_{33} = q_W + O(\varepsilon^{3/2}). \tag{295}$$

Choose $\varepsilon_1 > 0$ small enough that the absolute remainders in these three equations are bounded by $c_1\varepsilon^2/2$, $c_2\varepsilon/2$, and $(1 - q_W)/2$, respectively, whenever $0 < \varepsilon \le \varepsilon_1$. Then choose

$$0 < \varepsilon_0 \le \min\left\{\varepsilon_1, 1, \overline{\Gamma}^{-1}, \bar{\eta}_{\text{drift}}, \frac{c_2}{c_1}, \sqrt{\frac{1 - q_W}{c_1}}\right\}. \tag{296}$$

For $0 < \varepsilon \le \varepsilon_0$, all three ratios are at most

$$q_\varepsilon = 1 - \frac{c_1}{2}\varepsilon^2 : \qquad 1 - \frac{c_2}{2}\varepsilon \le q_\varepsilon, \qquad \frac{1 + q_W}{2} \le q_\varepsilon.$$

Since $0 < c_1 < 1/2$ and $\varepsilon \le 1$, this number belongs to $(0, 1)$. This proves (87); (296) also enforces all the basic step bounds. □

Proof of Theorem 5.20. By (89) and Theorem E.31, the initial target $X_0^\sharp = X_\rho^\sharp(y_0, P_0)$ exists uniquely. Thus $\xi_0 = \left\|X_0 - X_0^\sharp\right\|_F^2$ and $\mathcal{L}_0$ are well defined. Put $e_0 = (d_0, \sigma_0, \xi_0)^\top$. All three coordinates are nonnegative, so (90) gives

$$e_{0,i} \le \frac{\mathcal{L}_0}{w_i} \le R_0 \frac{w_{\min}}{w_i} \le R_0 \qquad (i = 1, 2, 3). \tag{297}$$

We prove by induction that the iterates are well defined, the target $X_k^\sharp$ exists uniquely, and $\mathcal{L}_k \le q^k \mathcal{L}_0 \le R_0 w_{\min}$ for every $k \ge 0$. The initial case has just been established. Suppose this claim holds at iteration k, and write $e_k = (d_k, \sigma_k, \xi_k)^\top$. Positivity of the weights again gives

$$e_{k,i} \le \frac{\mathcal{L}_k}{w_i} \le R_0 \frac{w_{\min}}{w_i} \le R_0 \qquad (i = 1, 2, 3). \tag{298}$$

All the hypotheses of Theorem 5.18 are therefore valid at iteration k. That theorem ensures that the next updates are well defined and proves existence and uniqueness of $X_{k+1}^\sharp$ before defining ξ_{k+1}; in particular, (286) verifies the tracking condition at $k + 1$. It then gives $0 \le e_{k+1} \le M e_k$. Multiplication by $w^\top > 0$ and (88) yield

$$\mathcal{L}_{k+1} \le w^\top M e_k \le q w^\top e_k = q\mathcal{L}_k \le q^{k+1}\mathcal{L}_0 \le R_0 w_{\min}. \tag{299}$$

This closes the induction, proving existence and uniqueness of $X_k^\sharp$ for every $k \geq 0$ together with (92).

Since $w_i e_{k,i} \leq \mathcal{L}_k$, the three squared errors satisfy

$$d_k \leq \frac{q^k \mathcal{L}_0}{w_1}, \qquad \sigma_k \leq \frac{q^k \mathcal{L}_0}{w_2}, \qquad \xi_k \leq \frac{q^k \mathcal{L}_0}{w_3}.$$

The first bound gives $\|y_k - y_\star\|_2 = O(q^{k/2})$. Moreover, (298) and (82) imply $\sqrt{d_k} + \sqrt{\sigma_k} < r_{\text{track}}$ at every iteration. Applying Theorem E.31 and the triangle inequality gives

$$\begin{aligned} \|X_k - X_\star\|_F &\leq \left\|X_k - X_k^\sharp\right\|_F + \left\|X_k^\sharp - X_\star\right\|_F \\ &\leq \sqrt{\xi_k} + L_{X,1}(1 + C_P)(\sqrt{d_k} + \sqrt{\sigma_k}) = O(q^{k/2}). \end{aligned}$$

This proves (93). □

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